Wavefronts, Huygens principle & optical path

Constant-phase surfaces, secondary wavelet envelopes, Snell law proof, and optical path equivalence.

Wavefront shapes & ray orthogonality

A wavefront is a locus of points oscillating with equal phase at instant t. In an isotropic medium, energy rays travel everywhere perpendicular to wavefronts, while amplitude decays with area growth.

$$I_{\text{sph}} \propto \tfrac{1}{r^2}\ (A \propto r^{-1})$$

$$I_{\text{cyl}} \propto \tfrac{1}{r}\ (A \propto r^{-1/2}), \qquad I_{\text{plane}} = \text{const}$$

WHY / Total power crossing concentric spheres 4 pi r^2 or cylinders 2 pi r L stays constant in a lossless medium, forcing A ~ r^(-1) and r^(-1/2).

WATCH / In anisotropic crystals (calcite e-ray), energy walk-off separates Poynting vector S from wavevector k, so the e-ray is NOT perpendicular to its wavefront.

Huygens secondary wavelets & Snell law

Every point on primary wavefront AB emits spherical secondary wavelets spreading at local phase speed v; the new wavefront CD after dt is their forward tangent envelope.

$$\frac{\sin i}{\sin r} = \frac{BC/AC}{AD/AC} = \frac{v_1 \Delta t}{v_2 \Delta t} = \frac{v_1}{v_2} = \frac{n_2}{n_1} \Rightarrow n_1 \sin i = n_2 \sin r$$

WHY / Kirchhoff obliquity factor K(theta) = (1 + cos theta)/2 equals 1 forward (theta = 0) and 0 backward (theta = pi), explaining why no backward wave forms.

WATCH / Entering glass (n2 > n1) slows wavelets down (v2 < v1) and shortens wavelength lam = lam0/n, while frequency nu remains strictly unchanged.

Optical path length (OPL) & Fermat rule

Geometric distance x inside refractive index n accumulates phase k x = (2 pi n / lam0) x, identical to travelling vacuum distance n x. Inserting slab (mu, t) into air adds OPL (mu - 1) t.

$$\Delta_{\text{opt}} = \sum_k n_k x_k = \int n(\mathbf{r})\, ds, \qquad \Delta\phi = \tfrac{2\pi}{\lambda_0}\,\Delta_{\text{opt}} + \Delta\phi_{\text{refl}}$$

WHY / By Fermat principle, all rays from object O to aberration-free image I take equal time: an ideal lens converts angle theta to y = f tan(theta) with zero extra phase difference.

WATCH / Always divide Delta_opt by vacuum wavelength lam0! Dividing Delta_opt by medium wavelength lam0/n double-counts refractive index n.

Wavefront tilt & relativistic Doppler shift

A thin prism (A, mu) delays its base ray by (mu - 1) H A relative to its apex ray, tilting an incident plane wavefront by deflection angle delta = (mu - 1) A without altering spacing.

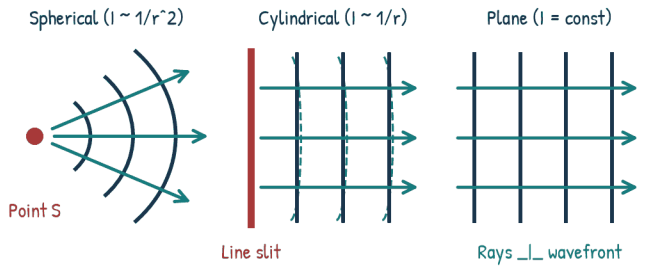
$$\frac{\Delta\nu}{\nu} = -\frac{\Delta\lambda}{\lambda} = \pm\frac{v_{\text{rad}}}{c}$$

$$(+ \text{ approach blueshift}, \quad - \text{ recession redshift})$$

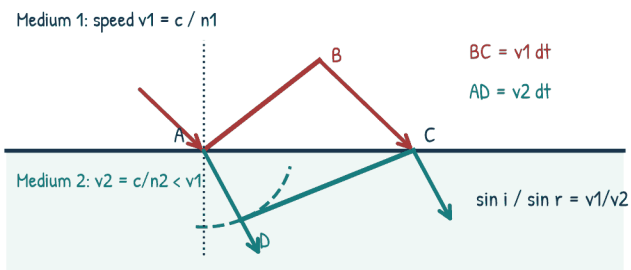
Worked check: H-alpha 656.3 nm measured at 656.96 nm (+0.66 nm shift) gives galaxy recession speed v = c (0.66 / 656.3) = 300 km/s.

WATCH / Colour sensation depends on photon energy h nu (invariant in water/glass), whereas astronomical redshift changes observed frequency nu itself.

Spherical, cylindrical and plane wavefronts with orthogonal energy rays



Huygens wavelet envelope proof of Snell refraction n1 sin i = n2 sin r



REMEMBER / Frequency nu is fixed by the source across any stationary boundary; OPL = sum(n_i x_i) always pairs with vacuum lambda0.

Superposition, phasors, visibility & coherence

Two-wave harmonic phasor sum, global power conservation, Michelson contrast, and wavefront vs amplitude splitting.

Phasor triangle & resultant irradiance

Two co-polarised harmonic fields of intensities I1, I2 and phase difference dphi add as rotating vectors of lengths sqrt(I1) and sqrt(I2) separated by exterior angle dphi:

$$I(\Delta\phi) = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi$$

$$I_{1=2}(\Delta\phi) = 4I_0\cos^2\left(\frac{\Delta\phi}{2}\right)$$

WHY / Constructive peaks reach (sqrt I1 + sqrt I2)^2 while destructive troughs drop to (sqrt I1 - sqrt I2)^2; equal beams swing from 0 to 4 I0.

WATCH / At path difference lam/6 (dphi = pi/3), cos^2(pi/6) = 3/4 gives I = 3 I0! Forgetting the half-angle and taking cos^2(pi/3) gives I0 by mistake.

Energy conservation & fringe visibility V

Dark fringes never destroy energy: spatial averaging <cos dphi> = 0 shifts missing energy from minima straight into maxima, keeping average irradiance strictly I1 + I2.

$$\langle I\rangle = \frac{I_{\max}+I_{\min}}{2} = I_1 + I_2, \qquad V = \frac{I_{\max}-I_{\min}}{I_{\max}+I_{\min}} = \frac{2\sqrt{\eta}}{\eta+1} \quad (\eta = \frac{I_1}{I_2})$$

Fast ratio anchors: amplitude ratio r_A = 3:1 means intensity ratio eta = 9:1, so I_max : I_min = (3+1)^2 : (3-1)^2 = 4:1 and visibility V = 0.60.

WATCH / Never confuse amplitude ratio A1:A2 with intensity ratio I1:I2 = A1^2 : A2^2 — misreading "amplitude ratio 2:1" yields 3+2sqrt(2) instead of 9:1!

Coherence & orthogonal polarisation law

Two independent lamps scramble phase randomly every ~1 ns; detectors average <cos dphi(t)> = 0 and record flat incoherent sum I1 + I2 with zero fringes (V = 0).

$$I(\mathbf{r}) = I_1 + I_2 + 2\sqrt{I_1 I_2}\,(\hat{\mathbf{e}}_1\cdot\hat{\mathbf{e}}_2)\cos\Delta\phi$$

WHY / By Fresnel-Arago laws, orthogonal polarisations (e1 . e2 = 0) never form intensity fringes, instead cycling linearly -> circularly -> linearly across space.

WATCH / Placing a 45-deg linear analyser AFTER perpendicular slit polaroids projects both beams onto one axis and immediately brings sharp fringes back!

Wavefront division vs amplitude division

Every interferometer derives coherent twin waves from ONE source by tapping two lateral patches of one wavefront OR partially reflecting one ray at an interface.

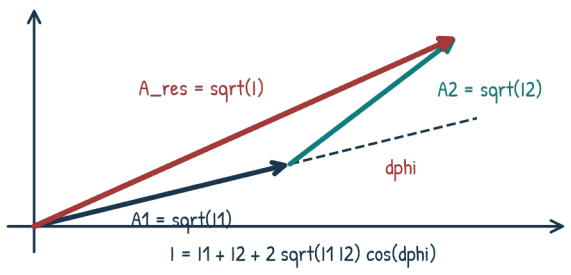
Wavefront : YDSE, Biprism, Lloyd, Billet

Amplitude : Films, Wedges, Newton, Michelson

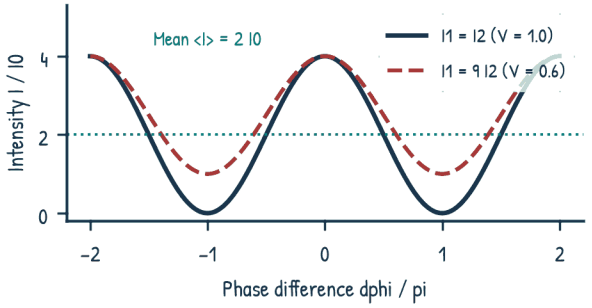
WHY / Wavefront splitters require a tiny spatial pinhole/slit (s << lam L1/d), whereas amplitude splitters work outdoors under the broad diffuse sky!

WATCH / Widening primary slit S in YDSE washes out fringes completely, while widening the sodium lamp above Newton rings makes the rings brighter AND sharper!

Rotating phasor triangle adding two co-polarised harmonic waves



Fringe contrast drops when unequal beam intensities leave minima non-zero



REMEMBER / In I = 4 I0 cos^2(dphi/2), use HALF the phase difference dphi/2; closing one slit drops central brightness from I_max to I_max/4.

Young double-slit experiment (YDSE) core laws

Exact path difference vs paraxial approximation, bright/dark indexing, angular width, and finite-order counting.

Exact geometry vs two-stage paraxial limit

Slits S1, S2 separated by d illuminate a parallel screen at distance D. Keep the two approximations distinct: D >> d gives parallel rays d sin(theta); |y| << D linearises sin(theta) ~ y/D.

$$\Delta x = \sqrt{D^2 + (y + \tfrac{d}{2})^2} - \sqrt{D^2 + (y - \tfrac{d}{2})^2} \approx d\sin\theta \approx \tfrac{dy}{D}$$

WHY / Even if y is several metres on a distant screen (theta = 45 deg), D >> d STILL holds accurately, so d sin(theta) = n lam stays exact while dy/D fails!

WATCH / When JEE gives d = 2 lam or d = 3 lam, dy/D puts n = 2 at y = 2D instead of infinity! Compute theta_n = arcsin(n lam / d) and y_n = D tan(theta_n).

Bright/dark positions, linear & angular beta

In the small-angle regime, bright and dark fringes alternate with equal linear spacing beta; angular spacing dtheta = beta / D depends ONLY on lam / d, independent of D.

$$y_{\text{bright}} = \tfrac{n\lambda D}{d} \ (n = 0, \pm 1, \ \dots)$$

$$y_{\text{dark}} = (n - \tfrac{1}{2})\tfrac{\lambda D}{d} \ (n = 1, 2, \ \dots), \qquad \beta = \tfrac{\lambda D}{d}$$

WHY / Moving the screen from D to 2D spreads linear spacing 2 beta in proportion to ray divergence, keeping angular separation lam / d invariant.

WATCH / Off-by-one classic trap: 5th dark fringe from central bright is ALWAYS (5 - 0.5) beta = 4.5 beta, NEVER 5.5 beta (check n = 1 gives 0.5 beta!).

Total maxima & minima on an infinite screen

Since |sin theta| <= 1 bounds |d sin theta| <= d, only finitely many interference orders exist. Check carefully whether integer endpoint d/lam lands at 90 deg (y = +/- infinity)!

$$N_{\text{bright, hemisphere}} = 2\lfloor \tfrac{d}{\lambda} \rfloor + 1, \qquad N_{\text{dark, hemisphere}} = 2\lfloor \tfrac{d}{\lambda} + \tfrac{1}{2} \rfloor$$

Concrete check: d = 4.3 lam gives 9 bright and 8 dark; d = 4.7 lam gives 9 bright and 10 dark; exact d = 4.0 lam gives 7 finite hits on a flat parallel screen (9 including +/-90 deg).

WATCH / Always check whether the stem asks for fringes "on a large flat screen parallel to slits" (|theta| < 90 deg strict!) vs "in all directions in front of slits".

Irradiance profile & single-slit closure anchor

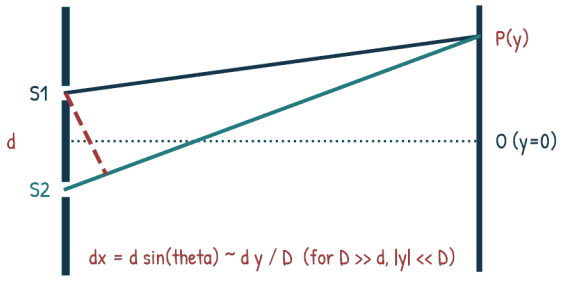
Within one fringe cycle [0, beta], intensity drops smoothly: y = 0 -> 4 I0; y = beta/6 -> 3 I0; y = beta/4 -> 2 I0 (half-maximum); y = beta/3 -> I0; y = beta/2 -> 0.

$$I(y) = 4I_0\cos^2\left(\tfrac{\pi y}{\beta}\right), \qquad I_{\text{one slit closed}} = I_0 = \tfrac{I_{\text{central}}}{4}$$

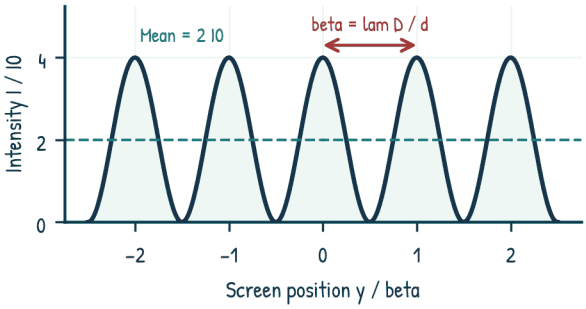
Worked check: d = 0.5 mm, D = 1.0 m, lam = 600 nm -> beta = 1.2 mm. At y = 0.2 mm = beta/6, I = (3/4) I_central; blocking S2 drops 0 to I_central / 4!

WATCH / When problem defines I_max = I (not I0!), closing one slit drops intensity at 0 to I / 4 (25%), NEVER I / 2 — read what symbol I represents!

Young double-slit geometry and path difference dx = S2P - S1P



Equal-intensity YDSE profile redistributes energy without loss (<I> = 2 I0)



REMEMBER / Use y_n = n lam D / d ONLY when lam << d; when d is a few wavelengths, solve sin(theta_n) = n lam / d first and put y_n = D tan(theta_n).

3D hyperboloids, end-on circular rings & screen tilts

Full 3D locus around two point sources, perpendicular circular YDSE, full-circle spot counts, and bichromatic beats.

3D hyperboloid shells around two point sources

For point sources S1, S2 separated by d, locus |r1 - r2| = n lam (< d) forms a family of two-sheeted hyperboloids of revolution sharing foci S1 and S2; n = 0 is the bisector plane.

$$\frac{x^2}{(n\lambda/2)^2} - \frac{y^2+z^2}{(d/2)^2-(n\lambda/2)^2} = 1 \qquad (0 < |n|\lambda < d)$$

WHY / Slicing parallel to S1-S2 at z = D gives hyperbolas that look straight near O; replacing pinholes by long parallel slits makes them strictly straight lines.

WATCH / In JEE multi-correct questions, point pinholes on a flat parallel screen form hyperbolas, NOT circles and NOT strictly straight lines far off axis!

End-on screen (S1 S2 _|_ screen) -> circular rings!

Align S2-S1 along the normal to screen at distance D >> d. Rotational symmetry around axis S2-S1-O slices the hyperboloids into concentric circular fringes!

$$\Delta x(\theta) = d\cos\theta \approx d\left(1 - \frac{r^2}{2D^2}\right) = m\lambda \Rightarrow r_p = D\sqrt{\frac{2p\lambda}{d}}$$

WHY / At centre O (theta = 0), path difference hits MAXIMUM d (order m0 = d/lam), decreasing outward as r grows — exact opposite of parallel YDSE!

WATCH / If d shrinks slowly in end-on YDSE, central order d/lam drops and circular fringes collapse INWARD into O and vanish one every delta_d = lam!

Tilted screen & full 360-deg circle spot counts

Tilting the viewing screen by angle psi stretches linear fringe width along the screen by 1 / cos(psi) while angular fringe width dtheta = lam / d stays unchanged.

$$\beta_{\text{tilt}} = \frac{\beta}{\cos\psi}, \qquad N_{360^\circ} = 4k + 2 \quad (d/\lambda \notin \mathbb{Z})$$

$$N_{360^\circ} = 4k \quad (\text{when } d = k\lambda \text{ exact: endfire merge!})$$

WHY / Around a full circle in a plane containing S1, S2: when d = k lam exact, end-fire spots at 0 deg and 180 deg each merge two symmetric branches into ONE point!

WATCH / Classic JEE Advanced integer question: d = 3.0 lam around a 360-deg circle gives 4(3) = 12 maxima, whereas d = 3.2 lam gives 4(3) + 2 = 14 maxima!

Bichromatic YDSE coincidence & washout

Superpose discrete wavelengths lam1 and lam2: bright fringes coincide at y = 0 and repeat periodically wherever integer multiples match n1 lam1 = n2 lam2.

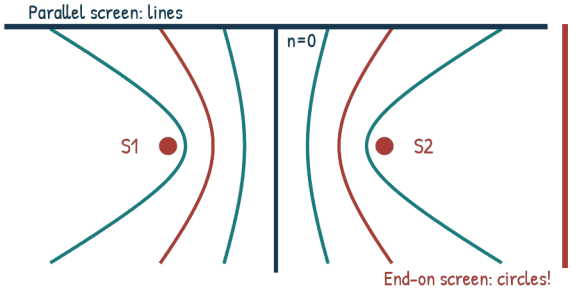
$$y_{\text{coinc}} = \text{LCM}(\beta_1, \beta_2) = \frac{q\lambda_1 D}{d} = \frac{p\lambda_2 D}{d}$$

$$\left(\frac{\lambda_1}{\lambda_2} = \frac{p}{q} \text{ in lowest terms}\right)$$

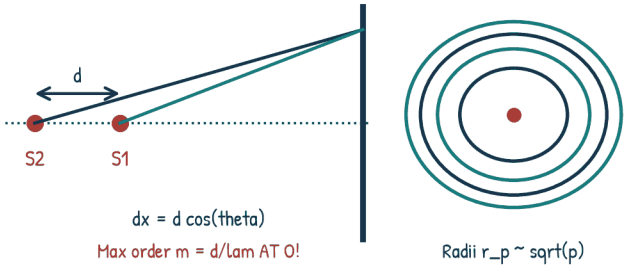
Check: 600 nm & 480 nm (ratio 5:4) -> 4th bright of 600 nm meets 5th bright of 480 nm every 4 beta_600! Notice odd/even parity: (2k-1)/(2n) = 5/4 has ZERO solution -> never washes to zero!

WATCH / Shorter wavelength ALWAYS has the LARGER integer order at coincidence (n_blue > n_red); check parity before claiming complete dark-on-bright washout exists!

Constant path-difference hyperboloids around S1-S2 sliced two ways



End-on screen perpendicular to S1-S2 produces circular fringes



REMEMBER / Parallel screen slices hyperboloids into straight lines (order 0 at O); end-on screen slices them into circles (MAX order d/lam at O!).

YDSE perturbations I: slabs, liquids & slit widths

Single/dual dielectric plates, liquid immersion, slab-inside-liquid shifts, unequal slit widths, and grey filters.

Dielectric plates over one or both slits

Placing plates (mu1, t1) on S1 and (mu2, t2) on S2 adds optical delay (mu1 - 1)t1 - (mu2 - 1)t2, shifting the zero-phase central fringe WITHOUT changing fringe width beta:

$$\Delta y = \frac{D}{d}[(\mu_1 - 1)t_1 - (\mu_2 - 1)t_2], \qquad N_{\text{shift}} = \frac{|\Delta y|}{\beta} = \frac{|\Delta_{\text{slab}}|}{\lambda_0}$$

WHY / Moving upward toward S1 shortens S1P in air, compensating the extra phase delay inside S1 plate so total optical paths S1P and S2P match again.

WATCH / Exchanging plates (mu1, t) <-> (mu2, t) moves central bright from +dy to -dy: total screen shift equals TWICE one-way shift = 2 |mu1 - mu2| t D / d!

Full liquid immersion & slab inside liquid

Submerging YDSE into liquid mu_L shrinks wavelength to lam0 / mu_L. If slab (mu_S, t) covers S1 inside liquid mu_L, the slab replaces thickness t of liquid (NOT air!):

$$\beta_L = \frac{\beta_{\text{air}}}{\mu_L}, \qquad \Delta y_{\text{in liquid}} = \frac{(\mu_S - \mu_L)t\,D}{\mu_L\,d}, \qquad N_{\text{shift}} = \frac{(\mu_S - \mu_L)t}{\lambda_0}$$

WHY / If mu_S < mu_L (e.g. hollow air cell mu_S = 1 submerged in water mu_L = 4/3!), phase advances in S1 so fringes shift AWAY from S1 toward S2!

WATCH / Separate two different setups: slab in air BEFORE slits + liquid between slits & screen gives dy = (mu_S - 1) t D / (mu_L d)! Write OPL = OPL every time!

Unequal slit widths & absorbing filters

In standard JEE/NCERT problems, transmitted slit irradiance scales linearly with narrow slit width (I1 : I2 = w1 : w2), unless the stem explicitly specifies amplitude A ~ w!

$$\frac{I_1}{I_2} = \frac{w_1}{w_2} \Rightarrow \frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{w_1} + \sqrt{w_2}}{\sqrt{w_1} - \sqrt{w_2}}\right)^2$$

Check w1 : w2 = 4 : 1 -> sqrt(I1):sqrt(I2) = 2:1 -> I_max : I_min = 9 : 1 (V = 0.80). If S1 carries a 25% transmission neutral filter, I1 = I0/4 gives I_max = (9/4) I0, I_min = I0/4!

WATCH / A lossless clear glass slab changes ONLY phase (I_min stays strictly 0!); an absorbing smoked glass plate reduces amplitude AND raises I_min > 0!

Intensity at geometric centre O after slab insertion

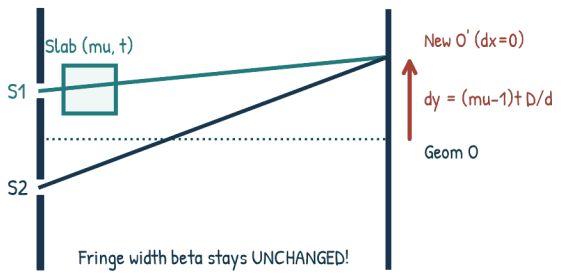
Inserting plate (mu, t) over S1 leaves phase at old centre O equal to dphi_O = 2 pi (mu - 1) t / lam0, cycling O periodically from 4 I0 -> 0 -> 4 I0 as t grows:

$$I(O) = 4I_0\cos^2\left(\frac{\pi(\mu - 1)t}{\lambda_0}\right) \Rightarrow t_{\min,\,\text{dark}} = \frac{\lambda_0}{2(\mu - 1)}$$

Quick check: lam0 = 600 nm, mu = 1.5 -> minimum thickness turning O completely dark is t = 600 / (2 x 0.5) = 600 nm; at t = 200 nm, dphi = pi/3 -> cos^2(pi/6) gives I(O) = 3 I0!

WATCH / Notice infinitely many thicknesses t = (2k - 1) lam0 / [2(mu - 1)] give identical monochrome pattern at O — white light breaks the degeneracy cleanly!

Dielectric plate over S1 shifts the zero-order fringe toward S1



Four high-frequency YDSE medium and slit-width perturbation rules

Full liquid immersion (mu_L) lam -> lam0 / mu_L beta_L = beta_air / mu_L	Slab mu_S inside liquid mu_L dy = (mu_S - mu_L) t D / (mu_L d) Shift N = (mu_S - mu_L) t / lam0
Unequal widths: I ~ w rule I1 : I2 = w1 : w2 Imax/Imin = (sqrt w1 + sqrt w2)^2 / ...	Plate swap between S1 & S2 Moves from +dy to -dy Total travel = 2 |mu1 - mu2| t D / d

REMEMBER / Fringes ALWAYS shift toward the slit with larger optical delay; fringe width beta stays unchanged in air and scales as beta/mu_L in liquid.

YDSE perturbations II: oblique rays, motion & white light

Off-axis sources, tilted incident waves, fringe velocities, and spectrometer missing wavelengths in white light.

Off-axis source & oblique plane incidence

Shifting primary slit S up by y_s (distance D1 before slits) OR tilting plane wave upward by alpha introduces a pre-slit phase bias that shifts the zero-order bright fringe:

$$y_c = -\frac{D}{D_1} y_s$$

or $\Delta_{\text{tot}} = d\sin\theta - d\sin\alpha = n\lambda \Rightarrow \theta_c = +\alpha$

WHY / Because |sin theta| < 1 on a parallel screen, allowed orders run unsymmetrically from -d(1 + sin alpha)/lam to +d(1 - sin alpha)/lam!

WATCH / At alpha = 30 deg with d = 2 lam: -3 < n < +1 leaves ONLY n = -2, -1, 0 (3 fringes on parallel screen)! Never apply symmetric 2 floor(d/lam) + 1 when alpha != 0!

Moving source, slits, screen & gas index rates

Differentiate y_n(t) directly with respect to time! Every moving-part YDSE question drops out cleanly without memorising separate formulas:

$$v_{\text{source shift}} = -\frac{D}{D_1} v_s$$

$$v_{\text{screen}} = \frac{n\lambda}{d}\dot{D} = \frac{y_n}{D}\dot{D}, \qquad \frac{dN}{dt} = \frac{L}{\lambda_0}\left|\frac{d\mu}{dt}\right|$$

WHY / Source or slab motion translates EVERY order rigidly at the same speed, whereas moving the screen keeps n = 0 pinned at 0 while outer orders fan out!

WATCH / Check sign of v_fringe: when source S moves UP (+y), central bright fringe moves DOWN (-y) by magnification ratio D / D1!

White-light YDSE & missing wavelengths at P

With white light (400-700 nm), y = 0 has dx = 0 for ALL wavelengths -> pure white central fringe, flanked by warm reddish inner edge (violet hits first dark minimum first) and 4-6 rainbow fringes.

$$\lambda_{\text{missing}}(k) = \frac{2\Delta x_P}{2k-1} \quad (k = 1, 2, 3, \ldots)$$

$$\lambda_{\text{bright}}(n) = \frac{\Delta x_P}{n} \quad (n = 1, 2, \ldots)$$

WHY / A spectroscope placed at fixed y_P sees a channeled spectrum crossed by jet-black absorption-like dark bands at every wavelength suffering destructive interference at P.

WATCH / Keep ONLY candidates strictly inside the problem visible window (usually 400 to 700 nm); discard IR (k = 1, 2) and UV lines immediately!

Point directly opposite S1 (y_P = d / 2) anchor

Standard JEE favourite: spectrometer slit sits at y_P = d/2 directly opposite slit S1 (so S1P = D and S2P = sqrt(D^2 + d^2)):

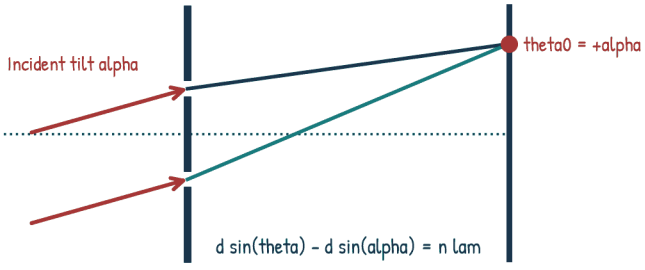
$$\Delta x_P = \sqrt{D^2 + d^2} - D \approx \frac{d^2}{2D}$$

$$\Rightarrow \lambda_{\text{missing}} = \frac{d^2}{D}, \ \frac{d^2}{3D}, \ \frac{d^2}{5D}, \ \ldots$$

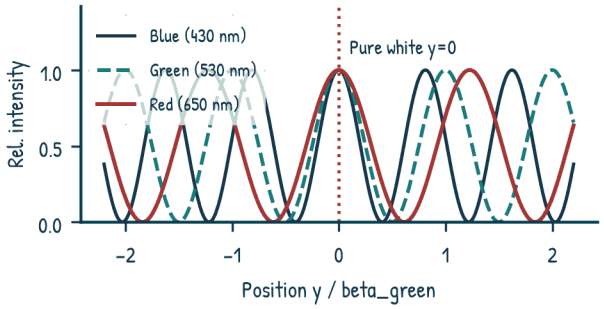
Check: d = 1.0 mm, D = 1.0 m -> d^2/D = 1000 nm (IR), d^2/(3D) = 333 nm (UV) -> ZERO missing visible lines unless d = 1.2 mm (gives 1440, 480 nm blue)!

WATCH / If D is NOT huge compared with d (e.g. D = 2d), NEVER use d^2/(2D): compute exact sqrt(5) d - 2 d = 0.2361 d directly!

Oblique plane incidence shifts central fringe to theta = alpha



Polychromatic YDSE: white central fringe and missing wavelengths at y_P



REMEMBER / Pre-slit path difference ADDS algebraically to post-slit path difference; missing wavelengths at P satisfy lam = 2 dx_P / (2k - 1).

Parallel thin films: phase flips & R + T = 1

Exact 2 mu t cos(r) derivation, Stokes reflection pi-jump rules, transmitted complementarity, and soap black film.

Exact geometric OPL = 2 mu t cos(r) proof

Inside film (mu, t) at refracted angle r (n1 sin i = mu sin r), internal ray travels 2 t / cos(r), while upper-surface reflected ray leads in ambient n1 by AN = 2 t tan(r) sin(i):

$$\Delta_{\text{geom}} = \frac{2\mu t}{\cos r} - 2t\tan r\,(\mu \sin r) = 2\mu t \cos r$$

$$\Delta_{\text{geom}} = 2t\sqrt{\mu^2 - n_1^2 \sin^2 i}$$

WHY / Tilting away from normal (i increases -> r increases -> cos r decreases) REDUCES 2 mu t cos(r), shifting thin-film reflected colours toward shorter blue wavelengths!

WATCH / Never write 2 mu t / cos(r)! Subtracting the air wavefront advance multiplies by (1 - sin^2 r) = cos^2 r, leaving cos(r) in the NUMERATOR!

Four-way phase jump table (zero guesswork)

Reflection from optically denser medium (n_from < n_to) flips phase by pi (lam0/2); reflection from rarer medium (n_from > n_to) flips phase by 0!

$$n_1 < \mu > n_3: \quad \text{Bright } (m - \tfrac{1}{2})\lambda_0, \quad \text{Dark } m\lambda_0$$

$$n_1 < \mu < n_3: \quad \text{Bright } m\lambda_0, \quad \text{Dark } (m - \tfrac{1}{2})\lambda_0$$

WATCH / Kerosene (1.45) on water (1.33) has ONE flip (pi - 0 = pi); MgF2 (1.38) OR water (1.33) on crown glass (1.52) has TWO flips (pi - pi = 0)!

Energy conservation R + T = 1 & contrast gap

For every lossless dielectric film, reflected and transmitted patterns are strictly complementary at every wavelength and viewing angle:

$$R(\lambda) + T(\lambda) = 1$$

$$\Rightarrow \text{Refl DARK } (R = 0) \Leftrightarrow \text{Trans MAX } (T = 100\%)$$

WHY / At air-soap (R_surf ~ 2%), the two reflected rays have equal intensities (~2% vs ~1.9%) giving 100% black contrast, whereas transmitted rays are 96% vs 0.04% (tiny 4% ripple)!

WATCH / If white light reflects green maximally at thickness t, transmitted light looks magenta (green depleted), NOT dark!

Draining soap film & ultra-thin black film

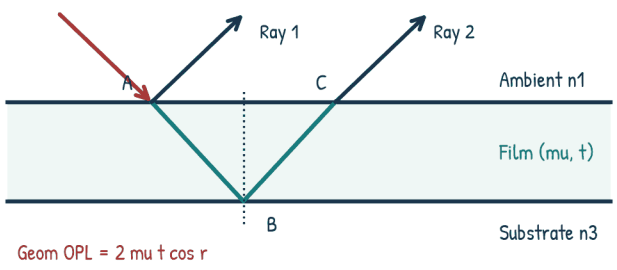
In a vertical soap film (mu ~ 1.34) in air, gravity thins the top and thickens the bottom: bottom (> 5 um) looks silvery white; middle shows vivid horizontal colour bands.

$$\lambda_{\text{peak, 1st}} = 4\mu t \;\to\; 0 \text{ as } t \to 0 \Rightarrow R(\lambda_{\text{vis}}) \to 0$$

WHY / Audit note (Wave-optics.md 8.5 Q9): as an oil/soap film thins below ~100 nm, peak reflected lam = 4 mu t MONOTONICALLY DECREASES (yellow -> violet -> black), never increases!

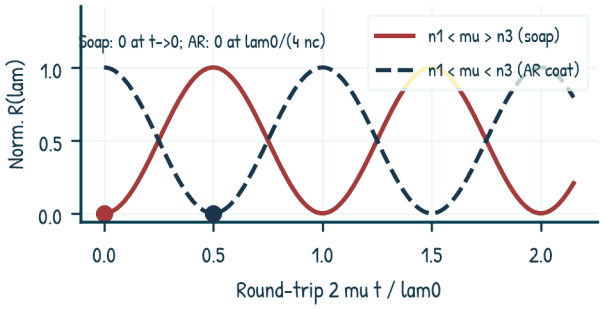
WATCH / A soap film looks jet black just before popping NOT because it absorbs light, but because t << lam/4 leaves ONLY the pi top-surface flip (T = 100%)!

Parallel thin film reflection geometry giving OPL = 2 mu t cos r



Geom OPL = 2 mu t cos r

Opposite vs matching boundary phase jumps swap bright and dark orders



REMEMBER / Compare film index mu against BOTH surrounding media n1 and n3 BEFORE writing m lam vs (m - 1/2) lam!

Wedge thin films & Newton circular rings

Equal-thickness Fizeau contours, wire metrology, Newton sagitta formula, dust immunity, lens lift, and curved bases.

Air & liquid wedge films (Fizeau contours)

Two flat glass plates touching at x = 0 with spacer thickness h at distance L enclose small angle theta ~ h/L (thickness t = x theta). Under normal incidence with mu < n_glass:

$$2\mu\, x_m\theta = m\lambda_0 \quad (\text{DARK},\ m = 0, 1, \ \dots)$$

$$\beta = \frac{\lambda_0}{2\mu\theta} = \frac{\lambda_0 L}{2\mu h}, \qquad N_{\text{tot}} = \frac{2\mu h}{\lambda_0}$$

WHY / Each fringe traces a constant-thickness contour t(x, y) = const: optical flats are tested to lam/20 flatness by checking whether wedge fringes are straight!

WATCH / Pressing gently on the upper glass plate narrows the gap: local thickness drops, so dark fringes shift TOWARD the thicker spacer end!

Newton rings & dust-proof difference formula

Plano-convex lens of large curvature radius R >> r on a flat glass plate forms air film sagitta t(r) = r^2 / (2R). In reflection with mu0 < n_glass, central contact spot is DARK:

$$D_{\text{dark},\,n} = \sqrt{\frac{4n\lambda_0 R}{\mu_0}}, \qquad \lambda_0 = \frac{\mu_0(D_{n+m}^2 - D_n^2)}{4mR}$$

WHY / If a dust speck or lens deformation adds unknown offset t0 at r = 0, D_n^2 shifts by -8 R t0 for EVERY ring identically and cancels cleanly in D_{n+m}^2 - D_n^2!

WATCH / Audit trap (Wave-optics.md 8.5 Q3): between 10th and 20th dark ring, m = 20 - 10 = 10 (NOT 20!); miscounting m halves your calculated wavelength!

Lens lift, intermediate-index oil & curved base

Three favourite JEE Advanced twists: (1) lifting lens by height h; (2) oil index between crown lens and flint plate (n_L < mu0 < n_P); (3) curved lower plate R2:

$$r_n^2 = R\left(\frac{n\lambda_0}{\mu_0} - 2h\right)$$

$$\frac{1}{R_{\text{eff}}} = \frac{1}{R_1} \pm \frac{1}{R_2} \quad (+\ \text{convex/convex},\ -\ \text{convex/concave})$$

WHY / As lens lifts (h increases), r_n shrinks: rings COLLAPSE INWARD into centre O and vanish one ring every dh = lam0 / (2 mu0)!

WATCH / With CS2 (mu0 = 1.63) between crown lens (1.52) and flint base (1.68), BOTH boundaries flip phase by pi -> CENTRAL SPOT TURNS BRIGHT!

Worked Newton rings comparison check

Setup: R = 1.0 m, lam0 = 600 nm, crown lens 1.52 on flint flat 1.68. Air gives 10th dark diameter D10 = sqrt(4 x 10 x 6e-7 x 1.0) = 4.90 mm.

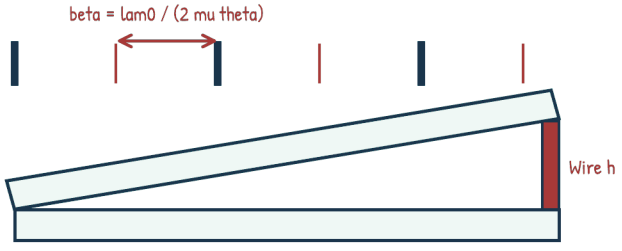
$$\text{Water } (4/3):\ D'_{10} = \frac{4.90}{\sqrt{4/3}} = 4.24 \text{ mm}$$

$$\text{Lift } h = 3.0\,\mu\text{m}:\ \Delta N = \frac{2h}{\lambda_0} = 10 \text{ rings}$$

Notice ring spacing shrinks outward: dr_n = r_{n+1} - r_n ~ lam0 R / (2 mu0 r_n) ~ 1 / sqrt(n) — outer rings crowd tightly together!

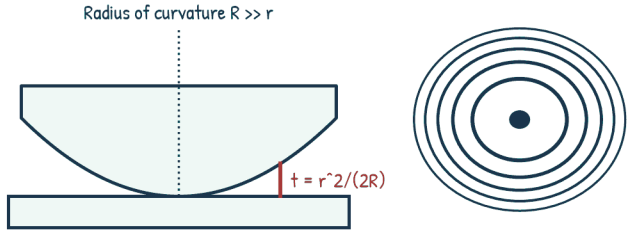
WATCH / In water (mu = 4/3), YDSE beta shrinks to 0.75x, whereas Newton diameter shrinks to sqrt(3)/2 = 0.866x — never apply 1/mu without the square root!

Air or liquid wedge film: equal-spacing fringes measure micron thickness h



Contact edge x = 0 is DARK; straight equal-thickness Fizeau lines

Newton rings: circular equal-thickness fringes crowd outerward as n^(-1/2)



D_n = sqrt(4 n lam0 R / mu0)

Dark centre (r = 0)

REMEMBER / Wedge fringes are equally spaced (beta = lam0 / 2 mu theta); Newton rings crowd outward as r_n ~ sqrt(n) and shrink by 1/sqrt(mu) in liquid!

AR coatings, Fresnel biprism, Lloyd & Billet splitters

Quarter-wave anti-reflection design, virtual-twin wavefront splitters, lens conjugate d = sqrt(d1 d2), and grazing pi proof.

Quarter-wave anti-reflection (AR) coatings

Coat glass n_g in air n1 with dielectric n_c (n1 < n_c < n_g). Both surfaces jump phase by pi (net jump 0), so destructive cancellation requires TWO simultaneous conditions:

$$t_{\min} = \frac{\lambda_0}{4 n_c} \quad (\text{odd multiples } (2m+1)\frac{\lambda_0}{4n_c})$$

$$n_c^{\text{ideal}} = \sqrt{n_1\, n_g}$$

WHY / Zeroing green 550 nm (t = 99.6 nm in MgF2, n_c = 1.38) leaves slight red (700 nm) + violet (400 nm) reflections, giving camera lenses their magenta sheen!

WATCH / Half-wave thickness t = lam0 / (2 n_c) is an ABSENTEE layer giving MAXIMUM reflection (same as bare uncoated glass), NEVER anti-reflection!

Fresnel biprism & conjugate lens d = sqrt(d1 d2)

Two acute prisms of tiny refracting angle A (in radians!) at distance a from slit S create virtual twins separated by d = 2 a (mu - 1) A, viewed at screen distance D = a + b:

$$d = 2a(\mu-1)A, \qquad \beta = \frac{\lambda(a+b)}{2a(\mu-1)A}, \qquad d = \sqrt{d_1\, d_2}$$

WHY / Sliding a convex lens (f < D/4) between biprism and screen gives two conjugate sharp image spacings d1 = (v/u) d and d2 = (u/v) d whose product eliminates magnification!

WATCH / Writing A = 0.5 instead of 0.5 x pi/180 = 8.73e-3 rad makes d too large by 57.3x! And only overlap zone W = 2 b (mu - 1) A carries interference fringes!

Lloyd single mirror & Fresnel double mirrors

Lloyd mirror places slit S height h above a front-surface mirror (d = 2h). Direct ray and grazing reflected ray overlap ONLY in upper half-screen y >= 0:

$$y_{\text{dark, Lloyd}} = \frac{m\lambda D}{2h} \quad (m = 0, 1, 2, \ \dots\ )$$

$$d_{\text{double mirror}} = 2r\sin\theta \approx 2r\theta$$

WHY / At contact edge y = 0 geometric path difference is zero yet intensity is JET BLACK — direct experimental proof of the pi reflection phase flip!

WATCH / Inserting thin mica (mu, t) in the direct ray of Lloyd mirror delays direct ray and pulls the pure white zero-phase bright fringe UP into view on screen!

Billet split lens real & virtual twin spacing

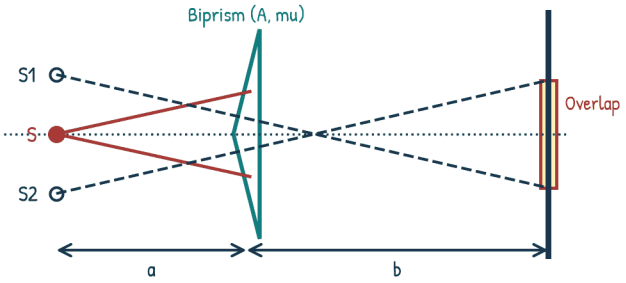
Cut convex lens f in half along a diameter and separate optical centres O1, O2 by gap eps. Point source S at u > f forms TWO real point images S1, S2 at v = u f / (u - f):

$$d_{\text{real}} = \frac{u+v}{u}\,\epsilon = (1+|m|)\,\epsilon, \qquad \beta = \frac{\lambda(L-v)}{d_{\text{real}}} \quad (L > u+v)$$

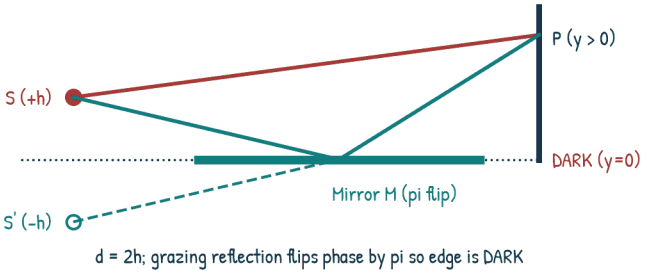
WHY / Chief rays from S through O1 (+eps/2) and O2 (-eps/2) diverge by angle eps/u and land at +/- (u + v)/u * (eps/2) in image plane z = u + v!

WATCH / Never set d = eps! At symmetric 2f setup (u = v = 2f), real twin separation is d = 2 eps (twice the lens cut gap)!

Fresnel biprism creates virtual twins d = 2a(mu - 1)A with A in radians



Lloyd mirror proves the reflection pi phase jump directly at contact line y = 0



d = 2h; grazing reflection flips phase by pi so edge is DARK

REMEMBER / Convert biprism/double-mirror angles from degrees to RADIANS first; Lloyd mirror has ONLY ONE grazing reflection so edge y = 0 is DARK.

Fraunhofer single-slit diffraction & phasor arc

Fresnel vs Fraunhofer regimes, destructive strip pairing, sinc^2 irradiance, 2x central lobe, and exact tan(beta) = beta peaks.

Fraunhofer condition & destructive strip pairing

Fraunhofer diffraction holds when rays are parallel (screen L >> a^2 / lam OR at focal plane y = f tan theta of a converging lens). Split slit width a into 2m equal zones:

$$a\sin\theta_{\text{dark}} = m\lambda$$

$$(m = \pm 1,\ \pm 2,\ \pm 3,\ \ldots\ ;\ \text{NEVER } m = 0!\ )$$

WHY / Zone k and zone k + m are separated by a/(2m); when (a/2m) sin theta = lam/2, every wavelet across the entire slit cancels its partner identically!

WATCH / Writing a sin(theta) = m lam for bright maxima by habit from YDSE is the #1 wave-optics exam mistake — m = 0 is the giant CENTRAL BRIGHT MAXIMUM!

Phasor-arc sinc^2 law & 2x central bright width

Summing infinitesimal slit strips across [-a/2, +a/2] bends a phasor chain of arc length A0 into circular arc subtending angle 2 beta, giving chord length A0 sin(beta)/beta:

$$I(\theta) = I_0\left(\frac{\sin\beta}{\beta}\right)^2 \quad \left(\beta = \frac{\pi a\sin\theta}{\lambda}\right), \qquad W_{\text{central}} = \frac{2\lambda D}{a} = 2W_{\text{side}}$$

WHY / Central bright peak spans from m = -1 to m = +1 (width 2 lam D / a), whereas each secondary bright lobe spans from m to m + 1 (width lam D / a)!

WATCH / Narrowing slit a SPREADS central fringe WIDER (W ~ 1/a)! Once a <= lam, first dark zero disappears beyond 90 deg and light fills the entire hemisphere!

Exact secondary peaks (tan beta = beta audit!)

Setting d/dbeta (sin beta / beta)^2 = 0 yields exact condition tan(beta) = beta, pulling secondary maxima slightly INWARD from (m + 1/2) pi and giving I_m / I0 = 1 / (1 + beta_m^2):

$$\beta_1 = 1.4303\pi \to \mathbf{4.72\%}$$

$$\beta_2 = 2.4590\pi \to \mathbf{1.65\%}, \qquad \beta_3 = 3.4709\pi \to \mathbf{0.83\%}$$

WHY / Audit fix (Wave-optics.md 5.9 Q6): |sin(2.4590 pi)| = 0.9917 gives (0.9917 / 7.7253)^2 = 1.65% (approx 4/(25 pi^2) = 1.62%), NOT sin(2.46 rad) = 0.66%!

WATCH / Over 90.3% of total transmitted slit power lands inside the central bright lobe alone — side peaks drop off more than 20x immediately!

Oblique incidence & lateral slit-shift immunity

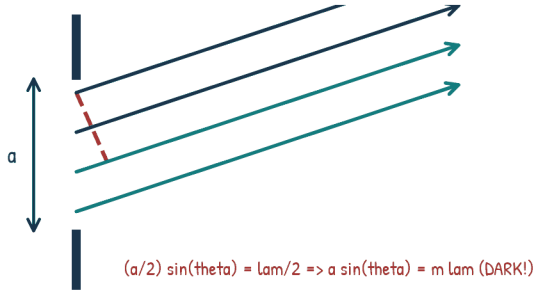
If plane wave hits slit a at angle alpha to normal, path difference across edges becomes a(sin theta - sin alpha), steering central maximum cleanly to theta_0 = +alpha:

$$a(\sin\theta_{\text{dark}} - \sin\alpha) = m\lambda \Rightarrow \Delta\theta_{\text{central}} \approx \frac{2\lambda}{a\cos\alpha}$$

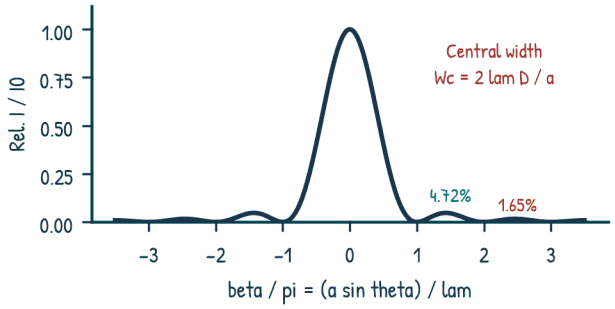
WHY / Placing a converging lens f after the slit maps angle theta uniquely to focal-plane coordinate y = f tan(theta), independent of lateral slit position!

WATCH / Sliding the single slit up or down in a parallel laser beam does NOT move the focal-plane diffraction pattern at all; moving the LENS shifts it!

Destructive zone pairing across a single slit of width a (m = +/-1, +/-2, ...)



Single-slit sinc^2 pattern: central lobe is twice as wide as side lobes



REMEMBER / In a single slit, a sin(theta) = m lam gives DARK MINIMA (m != 0), NEVER bright fringes; central bright lobe width is TWICE side-lobe width!

Double-slit diffraction envelope, missing orders & gratings

Product law for finite slits, exact visible-peak count 2 ceil(d/a) - 1, transmission gratings, and resolving power nN.

Finite double-slit product law & missing orders

Two slits of finite width a and centre-to-centre separation d (> a) multiply fast two-slit interference cos^2(alpha) by slow single-slit diffraction envelope sinc^2(beta):

$$I(\theta) = 4I_0\cos^2\left(\tfrac{\pi d\sin\theta}{\lambda}\right)\left(\tfrac{\sin(\pi a\sin\theta/\lambda)}{\pi a\sin\theta/\lambda}\right)^2, \qquad n_{\text{missing}} = m\,\tfrac{d}{a}$$

WHY / Whenever d sin(theta) = n lam coincides with envelope zero a sin(theta) = m lam, the n-th bright fringe has ZERO amplitude and vanishes completely!

WATCH / Check wording carefully: if slit width = a and opaque barrier = b, centre-to-centre spacing is d = a + b (so a = b means d/a = 2, even orders missing!).

Visible maxima inside central envelope (Q16 audit)

Central diffraction envelope runs strictly between -lam/a < sin(theta) < +lam/a, so interference order n has POSITIVE non-zero brightness iff |n| < d/a strictly:

$$N_{\text{central}} = 2\lceil d/a\rceil - 1 \;:\; \mathbf{2k-1}\quad (d/a = k\in\mathbb{Z})$$

$$N_{\text{central}} = \mathbf{2}\lfloor \mathbf{d/a}\rfloor + \mathbf{1}\quad (d/a\notin\mathbb{Z},\ \text{non missing edges})$$

WHY / Audit fix (Wave-optics.md Part 11 Q16): for a = 0.10 mm, d = 0.40 mm (d/a = 4), n = +/-4 hit zero envelope nodes -> STRICTLY 7 VISIBLE MAXIMA (-3..+3), NOT 9!

WATCH / Plug any number into 2 ceil(d/a) - 1: d/a = 4 -> 2(4) - 1 = 7; d/a = 4.2 -> 2(5) - 1 = 9! One universal formula handles integer AND fractional d/a!

Plane diffraction grating & angular dispersion

A grating of N rulings across width W has grating pitch d = a + b = W / N = 1 / n_lines. Constructive interference across all N slits sharpens lines to width ~ lam / (N d cos theta):

$$d\sin\theta_n = n\lambda, \qquad D_\theta \equiv \frac{d\theta}{d\lambda} = \frac{n}{d\cos\theta_n}, \qquad n_{\max} = \left\lfloor \tfrac{d}{\lambda}\right\rfloor$$

WHY / Peak intensity scales as N^2 I1 while line width shrinks as 1/N, concentrating all transmitted light into razor-sharp spectral lines separated by wide dark gaps!

WATCH / Convert rulings per cm to SI metres FIRST: "5000 lines/cm" means d = 10^(-2) / 5000 = 2.0e-6 m = 2000 nm (so at 600 nm, n_max = floor(2000/600) = 3)!

Grating resolving power R = nN & order overlap

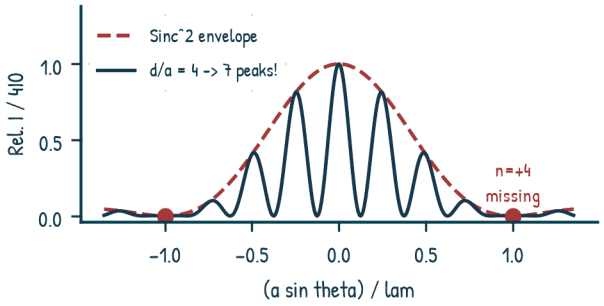
By Rayleigh criterion, lam + dlam is resolved in order n when its peak lands on the adjacent first zero of lam (path difference n N dlam = lam):

$$\mathcal{R} \equiv \frac{\lambda}{\Delta\lambda_{\min}} = n\,N_{\text{illuminated}}, \qquad \Delta\lambda_{\text{free}} = \frac{\lambda}{n}$$

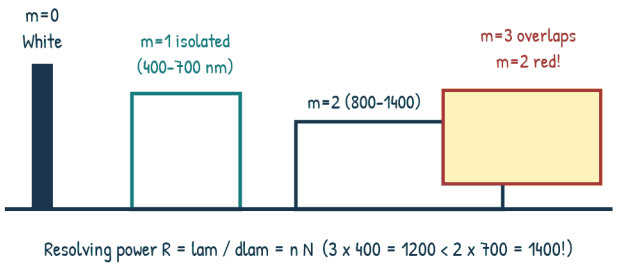
WHY / Dispersion dtheta/dlam depends ONLY on ruling spacing 1/d, whereas resolving power R = nN depends ONLY on TOTAL ILLUMINATED LINES N!

WATCH / In white light (400-700 nm), 2 x 700 = 1400 > 3 x 400 = 1200: 1st order is clean, whereas 2nd-order red ALWAYS overlaps 3rd-order violet!

Double slit (d/a = 4): 7 strictly visible bright peaks inside central lobe



Diffraction grating orders, chromatic resolving power nN, and order overlap



REMEMBER / For finite double slit, strictly visible bright peaks inside central envelope = 2 ceil(d/a) - 1 (7 for d/a = 4, NEVER 9!).

Airy disc, Rayleigh resolving power & pinhole optimum

Circular Bessel diffraction 1.22 lam/D, telescope & microscope limits, numerical aperture NA, and Fresnel distance Z_F.

Circular aperture Airy disc & factor 1.22

Integrating Huygens wavelets over a circular disk of diameter D gives Bessel amplitude 2 J1(u)/u instead of sinc, placing the first dark ring at u1 = 3.8317 = 1.220 pi:

$$\sin\theta_1 = \frac{1.22\,\lambda}{D}, \qquad r_{\text{Airy}} = \frac{1.22\,\lambda f}{D} = 1.22\,\lambda\,(f/\#)$$

WHY / Circular chords narrow near top and bottom rims, giving less high-offset weight than a square slit and widening the central bright disc from 1.00 to 1.22 lam/D.

WATCH / Watch radius R vs diameter D carefully: in terms of lens RADIUS R = D/2, theta_1 = 0.61 lam / R! Always check whether D or R is given!

Rayleigh criterion, telescopes & eye resolution

Two equal incoherent point stars are just resolved when peak 2 sits on Airy dark ring 1 (central saddle dips to 0.735 I_max, a 26.5% intensity dip):

$$\theta_{\min} = \frac{1.22\,\lambda}{D_{\text{obj}}}$$

$$\text{RP}_{\text{telescope}} = \frac{1}{\theta_{\min}} = \frac{D_{\text{obj}}}{1.22\,\lambda}, \qquad s_{\min} = L\,\theta_{\min}$$

Eye check (pupil D = 3 mm, lam = 550 nm): theta_min = 2.24e-4 rad -> two car headlights s = 1.5 m apart merge into one blur beyond L_max = s / theta_min = 6.7 km!

WATCH / Increasing eyepiece magnification M = f_o / f_e magnifies the Airy blur equally ("empty magnification") — ONLY larger objective D_obj sharpens resolution!

Microscope limit, oil immersion & electron waves

An objective collecting half-cone angle u through immersion medium mu has Numerical Aperture NA = mu sin(u) and resolves minimum specimen spacing d_min:

$$d_{\min} = \frac{1.22\,\lambda_0}{2\,\mu\sin u} = \frac{0.61\,\lambda_0}{\text{NA}}, \qquad \text{RP}_{\text{microscope}} = \frac{1}{d_{\min}} = \frac{2\,\mu\sin u}{1.22\,\lambda_0}$$

WHY / Cedarwood immersion oil (mu = 1.515) raises NA above 1.0 (to ~1.4), suppresses cover-slip TIR losses, and shrinks d_min down to ~200 nm in blue light.

WATCH / NCERT uses 1.22 lam0 / (2 mu sin beta) = 0.61 lam0/NA (self-luminous Rayleigh), whereas coherent Abbe questions use lam0 / (2 mu sin beta) = 0.50 lam0/NA — check options!

Fresnel distance Z_F & optimal pinhole diameter

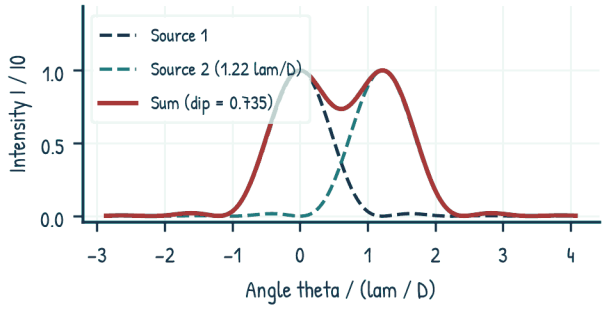
Ray optics stays valid up to Fresnel distance Z_F = a^2 / lam where diffraction spread equals beam width a. In a pinhole camera of length L, total blur radius R_tot is minimum at:

$$Z_F = \frac{a^2}{\lambda}, \qquad R_{\text{tot}}(D) = \frac{D}{2} + \frac{1.22\,\lambda L}{D} \Rightarrow D_{\text{opt}} = \sqrt{2.44\,\lambda L}$$

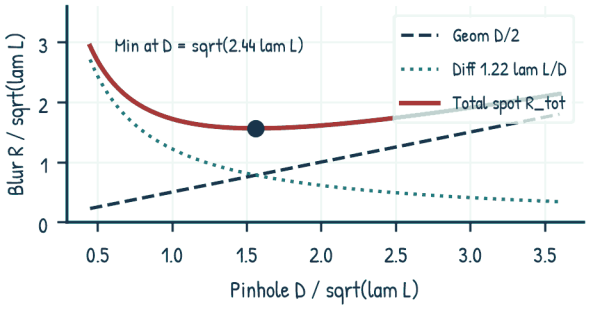
Numeric check: a = 2 mm, lam = 500 nm -> Z_F = 8.0 m. For pinhole box L = 20 cm, lam = 550 nm -> D_opt = sqrt(2.44 x 5.5e-7 x 0.20) = 0.52 mm!

WATCH / Shrinking a pinhole smaller than ~0.5 mm makes photos MORE blurry, not sharper, because diffraction spot 1.22 lam L / D explodes as 1 / D!

Circular Airy pattern and Rayleigh 26.5% central saddle resolution limit



Optimal pinhole diameter balances geometric blur against Airy diffraction



REMEMBER / 1D slit first zero is 1.00 lam/a while 2D circular aperture first dark ring is 1.22 lam/D; higher eyepiece M NEVER improves resolution!

Polarisation of light, Malus law & polaroid stacks

Transverse vector nature, unpolarised I0/2 theorem, crossed-polaroid middle sheet, and rotating-sheet harmonics.

Transverse waves & unpolarised I0 / 2 law

Polarisation proves light is strictly transverse (E _|_ k); longitudinal sound waves never polarise. Unpolarised light averages <cos^2 theta> = 1/2 across random azimuths:

$$I_{\text{after P}_1} = \frac{I_0}{2} \quad (\forall \text{ azimuth}), \qquad P = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{I_{\text{pol}}}{I_{\text{tot}}}$$

WHY / Orthogonal superpositions Ex = Ax cos(wt), Ey = Ay cos(wt + delta) trace lines at delta = 0, pi, circles when Ax = Ay and delta = +/-pi/2, and ellipses otherwise.

WATCH / Rotating ONE polaroid gives flat intensity I/2 for BOTH unpolarised AND circularly polarised light — insert a QWP first to tell them apart!

Malus law & 3-polaroid crossed-stack miracle

Linearly polarised light of intensity I1 hitting an ideal analyser tilted by theta passes amplitude A1 cos(theta), giving Malus law I2 = I1 cos^2(theta). With middle sheet P2(theta) between crossed P1(0) & P3(90):

$$I_{\text{unpol in}} = \frac{I_0}{2}\cos^2\theta \sin^2\theta = \frac{I_0}{8}\sin^2(2\theta)$$

$$I_{\text{pol in}} = \frac{I_{\text{pol}}}{4}\sin^2(2\theta)$$

WHY / Audit distinction (Wave-optics.md 12.6): peak at theta = 45 deg is I0 / 8 for unpolarised input before P1, and I_pol / 4 ONLY when input before P1 is already polarised along P1!

WATCH / Never write cos(theta) without squaring! At theta = 60 deg, cos(60) = 1/2 means AMPLITUDE halves while INTENSITY drops to cos^2(60) = 1/4!

Rotating polaroids & N-plate Optical Zeno limit

Spinning analyser P2 at angular velocity omega modulates intensity at 2 omega (4 peaks/rev); spinning middle sheet P2(omega t) between crossed sheets modulates at 4 omega (8 peaks/rev, mean I0/16)!

$$I_{N-\text{steps}} = \frac{I_0}{2}\left[\cos\left(\frac{\pi}{2N}\right)\right]^{2N} \to \frac{I_0}{2} \quad \text{as } N \to \infty$$

WHY / Each tiny step pi/(2N) loses O(1/N^2) intensity, so N steps lose N x O(1/N^2) -> 0: a continuous twisted stack rotates polarisation by 90 deg losslessly (LCD display principle)!

WATCH / For finite N = 2 steps across 90 deg (one 45-deg middle sheet + 90-deg sheet after P1): I_out = (I0/2)(cos 45)^4 = (I0/2)(1/4) = I0 / 8!

Partially polarised beam decomposition check

Every partially polarised beam equals an incoherent mixture of unpolarised intensity I_u plus linearly polarised intensity I_p (total intensity I_tot = I_u + I_p = I_max + I_min!):

$$I(\theta) = \frac{I_u}{2} + I_p\cos^2\theta \Rightarrow I_u = 2I_{\min}$$

$$I_p = I_{\max} - I_{\min}, \qquad I_{\text{tot}} = I_{\max} + I_{\min}$$

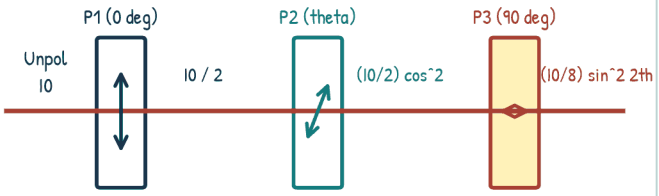
Worked check: analyser scan gives I_max : I_min = 5 : 1 -> degree of polarisation P = (5 - 1)/(5 + 1) = 2/3, and I_tot = 6 units (NOT 3)!

WATCH / Notice I_tot = I_max + I_min (SUM, not average!) because orthogonal polaroid positions pass I(0) + I(90) = I_tot!

Transverse electric-field trajectories for pure and natural polarisation states



Three-polaroid analyser train: rotated middle sheet restores blocked light



Peak output at theta = 45 deg: I0 / 8 (unpolarised) vs I_pol / 4 (polarised)

REMEMBER / Unpolarised light entering crossed polaroids with middle sheet at theta gives (I0/8) sin^2(2 theta); already-polarised input gives (I_pol/4) sin^2(2 theta)!

Brewster reflection law, critical angle & sky scattering

Dipole mechanism of tan(theta_B) = n2/n1, internal Brewster identity, pile of plates, and Rayleigh lam^(-4) scattering.

Brewster angle & oscillating dipole proof

Unpolarised light strikes dielectric boundary n1 -> n2 at Brewster angle theta_B when reflected and refracted rays are perpendicular (theta_B + theta_r = 90 deg):

$$\theta_B + \theta_r = 90^\circ \Rightarrow n_1 \sin\theta_B = n_2 \cos\theta_B \Rightarrow \tan\theta_B = \frac{n_2}{n_1}$$

WHY / In-plane p-dipoles inside medium 2 oscillate along the refracted ray perpendicular to E_t — pointing dead along the reflected ray where dipole radiation is IDENTICALLY ZERO!

WATCH / At theta_B, reflected light is 100% s-polarised (perpendicular to incidence plane), whereas refracted light is ONLY PARTIALLY p-polarised (~85% of s transmits)!

Internal Brewster angle & TIR shortcut identity

Going from denser medium n1 to rarer medium n2 ALSO has an internal Brewster angle tan(theta_B,int) = n2/n1 = 90 deg - theta_B,ext, ALWAYS strictly smaller than critical angle theta_c!

$$\theta_{B,\text{ext}} + \theta_{B,\text{int}} = 90^\circ$$

$$\sin\theta_c = \frac{n_2}{n_1} = \cot\theta_{B,\text{ext}} = \tan\theta_{B,\text{int}}$$

Standard benchmarks: air-water (4/3) gives theta_B = 53.13 deg (theta_r = 36.87 deg < theta_c = 48.75 deg); air-glass (1.5) gives theta_B = 56.31 deg (theta_r = 33.69 deg < 41.81 deg).

WATCH / Memorise sin(theta_c) = cot(theta_B): if a JEE question states "polarising angle in air is 60 deg", sin(theta_c) = cot(60) = 1/sqrt(3) instantly in 3 seconds!

Brewster laser windows, plate piles & road glare

Stacking 10-15 parallel glass plates at theta_B reflects away almost all s-light across 20-30 surfaces, leaving transmitted beam ~99% pure p-polarised with zero absorption loss!

$$\text{Horizontal water/asphalt surface}$$

$$\Rightarrow \text{reflected glare is HORIZONTALLY polarised } (s)$$

WHY / Gas laser tubes sealed with end windows tilted at theta_B give 0% reflection loss for p-waves and high reflection loss for s-waves, forcing pure linear laser output!

WATCH / Polaroid driving/fishing sunglasses ALWAYS align their pass axis VERTICALLY to extinguish horizontal road/lake reflection glare!

Rayleigh lam^(-4) sky scattering vs white clouds

Air molecules (size d ~ 0.3 nm << lam) scatter sunlight as driven Rayleigh dipoles with cross-section proportional to omega^4 ~ lam^(-4): blue 450 nm scatters 4.4x more strongly than red 650 nm!

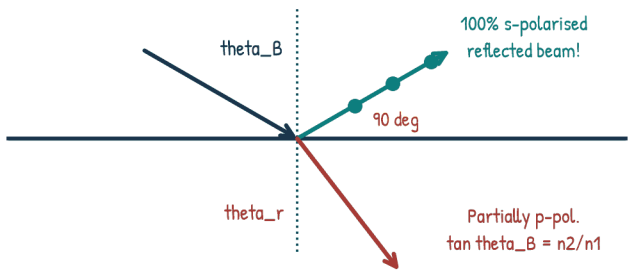
$$I_{\text{scat}}(\lambda, \Theta) \propto \frac{1 + \cos^2\Theta}{\lambda^4}$$

$$\Rightarrow \text{at } \Theta = 90^\circ, \text{ scattered blue light is linearly polarised!}$$

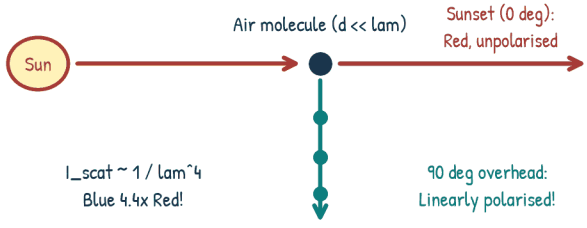
WHY / Looking 90 deg from the Sun, observer line-of-sight is perpendicular to solar rays, so ONLY transverse dipole oscillations perpendicular to the Sun-observer plane radiate to the eye!

WATCH / Cloud droplets (d ~ 10 um >> lam) scatter geometrically (~lam^0 -> white clouds!), while primary rainbows reflect near internal Brewster angle -> strongly polarised!

Brewster condition theta_B + theta_r = 90 deg zeroes reflected p-component



Rayleigh dipole scattering makes the sky blue and linearly polarised at 90 deg



REMEMBER / At Brewster angle ONLY the reflected beam is 100% linearly s-polarised; external Brewster & TIR critical angles satisfy sin(theta_c) = cot(theta_B)!

Double refraction, wave-plates & optical activity

Uniaxial crystals (o & e rays), Nicol prism, QWP & HWP retarders, QWP intensity audit fix, and Biot saccharimetry.

Uniaxial crystals: ordinary o vs extraordinary e

Calcite (CaCO3) and quartz split unpolarised light into two orthogonal linear rays: ordinary o-ray (speed c/n_o everywhere -> spherical wavelet, obeys Snell) and extraordinary e-ray (ellipsoidal wavelet)!

$$\text{Calcite } (-): \; n_e = 1.486 < n_o = 1.658 \; (v_e \geq v_o)$$

$$\text{Quartz } (+): \; n_e = 1.553 > n_o = 1.544 \; (v_e \leq v_o)$$

WHY / Along the optic axis direction ONLY, o and e wavelets touch tangentially (v_e = v_o = c/n_o), producing ZERO double refraction!

WATCH / Optic axis is a lattice DIRECTION throughout the crystal, NOT a single line! Nicol prism eliminates o-ray via TIR on Canada balsam (n_B = 1.55).

Quarter-wave (QWP) & half-wave (HWP) plates

A crystal plate cut with optic axis parallel to its faces accumulates phase retardation delta between orthogonal components E_o and E_e without absorbing any energy:

$$\delta = \tfrac{2\pi}{\lambda_0}\,|n_o - n_e|\,t \Rightarrow t_{\text{QWP}} = \frac{(2m+1)\lambda_0}{4\,|n_o - n_e|}, \qquad t_{\text{HWP}} = \frac{(2m+1)\lambda_0}{2\,|n_o - n_e|}$$

WHY / HWP mirrors linear polarisation across fast axis (rotating azimuth psi by 2 psi -> 90-deg flip at psi = 45 deg with ZERO energy loss!) and flips right <-> left circular!

WATCH / Linear light hitting a QWP at psi = 0 or 90 deg emerges STILL LINEAR! Circular light forms ONLY when |E_o| = |E_e|, i.e. strictly at psi = +/-45 deg!

QWP + analyser audit fix & 2-step mystery test

Audit fix (Wave-optics.md 6.9 Q7): send unpolarised I0 -> Polariser P1(0) -> lossless QWP(fast axis 45 deg) -> Analyser A(alpha):

$$I_0 \;\rightarrow\; \tfrac{I_0}{2}\;(\text{linear}) \;\rightarrow\; \tfrac{I_0}{2}\;(\text{circular}) \;\rightarrow\; \mathbf{\tfrac{I_0}{4}} \;\; \forall\,\alpha!$$

WHY / Circular light of total irradiance I0/2 has equal orthogonal components I0/4 + I0/4; ANY linear analyser passes ONE component = I0/4 and rejects the orthogonal I0/4!

WATCH / Mystery beam giving flat I/2 through rotating A: insert QWP BEFORE A! Circular -> turns linear (zeros at two angles!); Unpolarised -> stays flat I/2!

Optical activity (Biot law) vs Faraday rotation

Chiral media (quartz along optic axis, sugar solutions) have n_L != n_R, rotating linear polarisation without loss. In Biot saccharimetry, l is in DECIMETRES (1 dm = 10 cm) and c in g/cm^3:

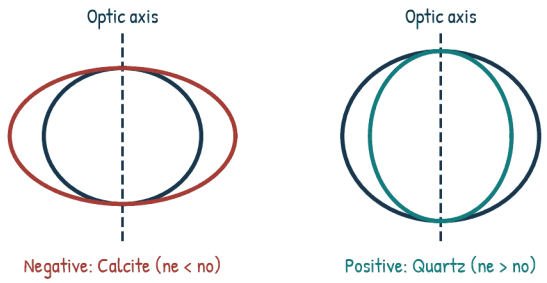
$$\theta = [\alpha]_\lambda^T\, l\, c, \qquad \alpha_{\text{rot}}(\lambda) \propto \lambda^{-2}$$

$$\text{Sucrose}: \; [\alpha]_D = +66.5^\circ\ \text{dm}^{-1}\,\text{g}^{-1}\,\text{cm}^3$$

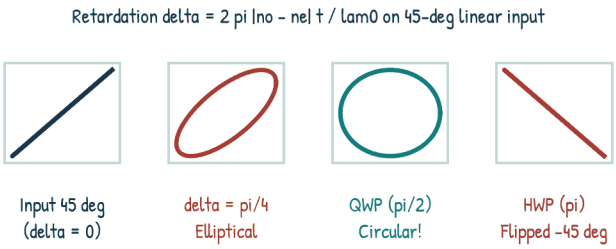
WHY / Reflecting light straight back through chiral sugar CANCELS net rotation (+theta - theta = 0), whereas magnetic Faraday rotation DOUBLES to 2 theta_F!

WATCH / UNIT TRAP: a 20 cm polarimeter tube has l = 2.0 dm (NOT 20 and NOT 0.2)! Always convert tube length to decimetres first!

Huygens spherical o-surface and ellipsoidal e-surface in uniaxial crystals



Evolution of polarisation state through quarter-wave and half-wave retarders



REMEMBER / Unpolarised I0 -> Polariser -> QWP(45 deg) -> Analyser ALWAYS outputs I0 / 4 at every analyser angle (NEVER I0 / 2)!

Wave optics master triage, scaling & exam checklist

Cross-topic formula discriminator, liquid/wavelength scaling table, numeric anchors, and top multi-correct traps.

Instant 6-setup formula discriminator

1. YDSE parallel: dy/D = n lam (bright), beta = lam D/d. 2. YDSE end-on: d cos(theta) = m lam (circles, max order at 0!). 3. Thin film: 2 mu t cos(r) + dense pi flips.

$$\text{Slit DARK}: \; a\sin\theta = m\lambda \;\; (m \neq 0)$$

$$\text{Airy DARK}: \; \theta_1 = \frac{1.22\lambda}{D}$$

$$\text{Grating BRIGHT}: \; d\sin\theta = n\lambda$$

4. Wedge dark: 2 mu x theta = m lam0. 5. Newton dark diameter: D_n = sqrt(4 n lam0 R / mu0). 6. Malus: I1 cos^2(theta); Brewster: tan(theta_B) = n2/n1, sin(theta_c) = cot(theta_B).

WATCH / Never confuse grating pitch d (BRIGHT at d sin theta = n lam) with single-slit width a (DARK at a sin theta = m lam)!

Liquid immersion & wavelength scaling test

Eliminate wrong algebraic options instantly by testing medium immersion (air -> liquid mu) and wavelength scaling (lam -> k lam):

$$\beta_{\text{YDSE}} \propto \frac{\lambda}{\mu}, \qquad W_{\text{slit}} \propto \frac{\lambda}{\mu}$$

$$D_{\text{Newton}} \propto \sqrt{\frac{\lambda}{\mu}}, \qquad D_{\text{pinhole, opt}} \propto \sqrt{\lambda L}$$

WHY / Newton sagitta is quadratic (t = r^2 / 2R), taking a SQUARE ROOT on lam/mu whereas every linear interference/diffraction angle scales to the FIRST power of lam/mu!

WATCH / Eye/camera Airy angle in outside air is STILL 1.22 lam0 / D_pupil because internal wavelength shrinkage lam0/n cancels Snell refraction n at the cornea!

High-yield numeric anchors worth memorising

Visible band: 400 nm (violet, 3.10 eV) to 700 nm (red, 1.77 eV), peak eye sensitivity 555 nm, hc = 1240 eV nm. Sodium D doublet: 589.0 & 589.6 nm (R_min = 982).

$$\text{Single slit peaks}: \; 1 : 0.0472 : 0.0165 : 0.0083$$

$$(\text{at } \beta/\pi = 0,\; 1.430,\; 2.459,\; 3.471)$$

Refractive indices: water 4/3 (theta_B = 53.1 deg), MgF2 1.38, crown glass 1.52 (R_normal = 4.3%), Canada balsam 1.55, calcite no = 1.658 / ne = 1.486, diamond 2.42.

WATCH / Rayleigh saddle intensity between two just-resolved stars is 0.811 I_max for 1D slit and 0.735 I_max for 2D circular Airy disc!

Top multi-correct exam traps summary

1. Closing one YDSE slit drops central intensity to I_max / 4. 2. Finite double slit with integer d/a = k has STRICTLY 2k - 1 central bright fringes (7 for d/a = 4).

3. Unpolarised -> crossed polaroids + 45-deg middle sheet gives I0 / 8; unpolarised -> P -> QWP(45) -> A gives I0 / 4. 4. Biprism angle A MUST be in radians!

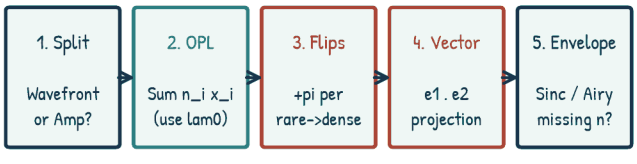
WHY / 5. Soap film t -> 0 looks BLACK in reflection and 100% BRIGHT in transmission; MgF2 on glass t -> 0 looks BRIGHT (two same-sign pi flips cancel!).

WATCH / Follow the 5-step phase checklist on every unfamiliar problem: Split -> OPL -> Dense pi flips -> Vector dot product -> Diffraction envelope!

Instant scaling comparison across interference, diffraction and polarisation

YDSE / Biprism beta = lam D / d (const) In liquid mu: beta / mu	**Newton's Rings** D_n = sqrt(4 n lam R / mu) In liquid mu: D_n / sqrt(mu)!
Slit vs Airy Disc Central slit = 2 lam D / a Airy dark = 1.22 lam / D	**Polaroid / QWP** Crossed + 45 deg -> I0 / 8 P + QWP(45) + A -> I0 / 4

Five-step phase bookkeeping checklist for zero sign or half-wave errors



Total dphi = (2 pi / lam0) d_opt + N_dense pi => Bright 2m pi, Dark (2m+1) pi

REMEMBER / Before writing = n lam, pause 2 seconds and name the setup: YDSE bright, thin-film dark, OR single-slit dark!

Michelson interferometer, Haidinger rings & FT spectroscopy

Double-pass arm geometry, dispersive compensator plate C, equal-inclination circular fringes, and doublet visibility beats.

Double-pass virtual air gap & compensator C

Beam splitter BS folds fixed mirror M2 onto virtual plane M2-prime separated by d = |L1 - L2| from movable mirror M1. Translating M1 by dx changes path difference by 2 dx:

$$\Delta_{\text{opt}}(\theta) = 2d\cos\theta = m\lambda \Rightarrow \Delta N = \frac{2\,\Delta x}{\lambda}$$

WHY / Without unsilvered compensator plate C, one arm crosses dispersive BS glass 3 times and the other once — d(lam) = 0 fails across colours and white-light fringes vanish!

WATCH / With identical plate C parallel to BS, one reflection is internal in glass (0) and the other external in air (pi): at exact d = 0 the central white-light fringe is DARK!

Haidinger circular rings vs Fizeau wedge lines

When M1 || M2-prime strictly, extended light produces circular equal-inclination Haidinger fringes localized at infinity; tilting M1 by alpha produces straight wedge fringes!

$$2d\left(1 - \frac{\theta_p^2}{2}\right) = (m_0 - p)\lambda \Rightarrow \theta_p = \sqrt{\frac{p\lambda}{d}}, \qquad \beta_{\text{tilted}} = \frac{\lambda}{2\alpha}$$

WHY / Exactly as in end-on circular YDSE, MAXIMUM order m0 = 2d/lam sits AT CENTRE theta = 0! Shrinking d makes circular rings collapse inward into 0!

WATCH / At d -> 0 one broad fringe fills the entire field ("infinite width"); slight tilt alpha straightens curved arcs into straight lines ONLY right at d = 0!

Gas refractometry & sodium D-doublet beats

Pumping gas into tube length L in one arm traverses the gas TWICE! With a spectral doublet (lam, lam + dlam), incoherent sum of two interferograms beats periodically to zero visibility:

$$n_{\text{gas}} - 1 = \frac{N\lambda_0}{2L}$$

$$V(\Delta x) = \left|\cos\left(\frac{2\pi\,\Delta\lambda\,\Delta x}{\lambda^2}\right)\right| \Rightarrow \Delta x_{\text{beat}} = \frac{\lambda^2}{2\,\Delta\lambda}$$

Sodium D anchor (589.0 & 589.6 nm, dlam = 0.6 nm): mirror travel between consecutive visibility washouts is dx_beat = (589.3e-9)^2 / (2 x 0.6e-9) = 0.289 mm!

WATCH / Watch factor of 2 carefully: mirror travel between washouts is lam^2 / (2 dlam), whereas round-trip optical path difference between washouts is lam^2 / dlam!

Michelson-Morley, LIGO & FTIR resolution

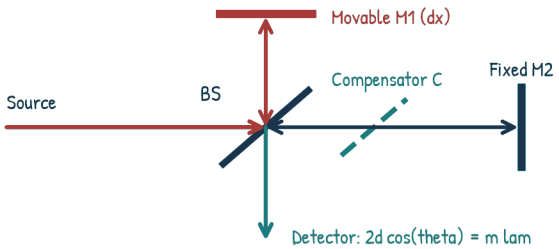
In Fourier-Transform (FTIR) spectroscopy, recording I(Delta) across total optical path scan Delta_max = 2 X_max recovers spectrum S(nu_tilde) with wavenumber resolution 1 / Delta_max:

$$\delta\tilde{\nu}_{\min} = \frac{1}{\Delta_{\max}} = \frac{1}{2X_{\max}} \Rightarrow \mathcal{R}_{\text{FTIR}} = \frac{\tilde{\nu}}{\delta\tilde{\nu}} = \frac{2X_{\max}}{\lambda}$$

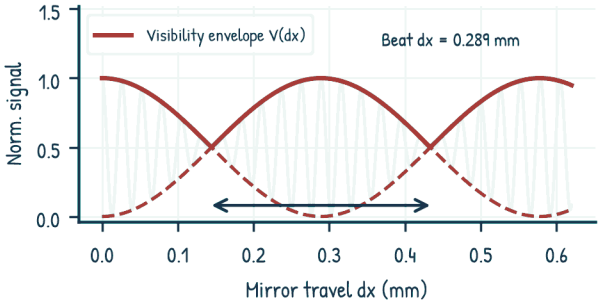
WHY / Michelson-Morley (1887) predicted 2 L v^2 / (lam c^2) ~ 0.4 fringe shift from Earth orbital motion and observed 0; LIGO uses 4 km arms to sense h ~ 10^(-21) strains!

WATCH / A compact 5 cm mirror scan (Delta_max = 10 cm) achieves resolving power R = 200,000 at 500 nm — matching a 200,000-line diffraction grating!

Michelson double-pass arms, virtual air plate d, and glass compensator plate C



Sodium D-doublet interferogram beats vanish every 0.289 mm of mirror travel



REMEMBER / Moving Michelson mirror M1 by dx shifts 2 dx / lam fringes (1 fringe per lam/2); gas cell length L enters TWICE as 2(n - 1)L!

Temporal & spatial coherence, group index & stellar baselines

Wiener-Khinchin Fourier duality, white-light group index shift, van Cittert-Zernike theorem, and Betelgeuse.

Temporal coherence & Wiener-Khinchin theorem

Equal-intensity fringe visibility V(tau) = |gamma(tau)| is the modulus of the Fourier transform of normalised power spectral density S(nu), fixing coherence time tau_c and length L_c:

$$\tau_c \sim \frac{1}{\Delta\nu}, \qquad L_c = c\,\tau_c \approx \frac{\lambda^2}{\Delta\lambda}, \qquad N_{\text{fringes}} \sim \frac{L_c}{\lambda} = \frac{\lambda}{\Delta\lambda}$$

WHY / Top-hat bandwidth gives |sinc(pi dnu tau)| (first zero at tau = 1/dnu); Lorentzian collision broadening gives exp(-pi dnu |tau|); Gaussian Doppler gives Gaussian decay.

WATCH / Sunlight has L_c ~ 1 um (~2 fringes!), Na discharge ~0.5 mm, low-pressure Kr-86 ~50 cm, and single-frequency He-Ne/fibre lasers reach L_c > 100 km!

Dispersion & white-light group-index zero shift!

Insert dispersive plate (n(lam), t) in one arm under WHITE light! Stationary phase d(dphi)/dk0 = 0 places the brightest white-light fringe at GROUP OPL, NOT phase OPL:

$$\Delta_{\text{air}}^{\text{white peak}} = (n_g - 1)\,t, \qquad n_g \equiv \frac{c}{v_g} = n(\lambda) - \lambda\,\frac{dn}{d\lambda}$$

WHY / In normal dispersion (dn/dlam < 0), group index n_g EXCEEDS phase index n! Individual monochrome fringes creep inside the envelope as t varies!

WATCH / IPhO classic trap: crown glass has n = 1.52 while n_g = 1.545 — reading the shifted achromatic peak directly measures n_g - 1 = 0.545 (5% higher than n - 1)!

Spatial coherence width & van Cittert-Zernike

An incoherent source of angular width theta_s = s / D1 washes out two-slit visibility according to the Fourier transform of the source brightness profile:

$$V_{\text{slit source}}(d) = \left|\operatorname{sinc}\!\left(\frac{\pi\, d\,\theta_s}{\lambda}\right)\right| \Rightarrow d_{\text{washout}} = \frac{\lambda}{\theta_s}, \qquad l_{\text{spatial}} \sim \frac{\lambda}{\theta_s}$$

WHY / Even completely independent atoms across a distant star become spatially coherent across baseline l_s ~ lam / theta_* simply by propagating far enough!

WATCH / Solar disc subtends theta_sun = 9.3e-3 rad -> transverse spatial coherence width of raw sunlight on Earth is ONLY ~60 um (why YDSE needs a pinhole first)!

Michelson 20-ft stellar interferometer (Betelgeuse)

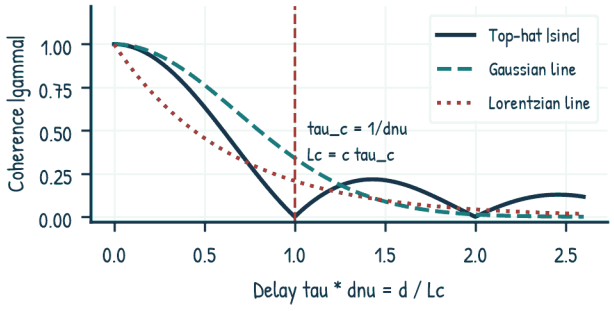
For a uniform circular star disc of angular diameter theta_*, 2D Fourier transform replaces sinc by Airy 2 J1(x)/x, driving fringe visibility to zero at outrigger baseline b:

$$V_{\text{star}}(b) = \left|\frac{2J_1(\pi b\,\theta_\star/\lambda)}{\pi b\,\theta_\star/\lambda}\right| = 0 \Rightarrow \theta_\star = \frac{1.22\,\lambda}{b_{\text{washout}}}$$

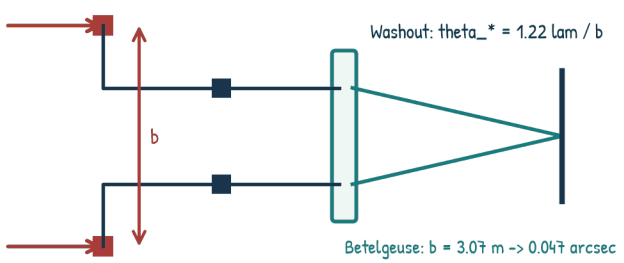
Historic 1920 Betelgeuse benchmark: lam = 575 nm, first fringe disappearance at b = 3.065 m -> theta_* = 2.29e-7 rad = 0.047 arcsec (~400 solar diameters at 640 ly)!

WATCH / Eyepiece fringe SPACING depends on FIXED inner mirror gap d_in (lam f / d_in), whereas fringe CONTRAST depends on outer periscope baseline b!

Wiener-Khinchin transform pairs spectral line shape with fringe visibility



Michelson periscope stellar interferometer beats atmospheric seeing limits



REMEMBER / Dispersive glass shifts the white-light zero fringe by GROUP index (n_g - 1)t, NOT phase index (n - 1)t! Stellar disc washes out at b theta_* = 1.22 lam.

Fresnel amplitude equations, Stokes laws & Bragg stacks

Exact TE (s) & TM (p) field coefficients, power conservation factor, incoherent thick-plate piles, and HR mirrors.

Exact Fresnel amplitude equations (s & p)

Matching tangential E_|| and H_|| across non-magnetic boundary n1 -> n2 (n1 sin i = n2 sin r) gives exact amplitude reflection and transmission ratios:

$$r_s = \frac{n_1\cos i - n_2\cos r}{n_1\cos i + n_2\cos r} = -\frac{\sin(i-r)}{\sin(i+r)}$$

$$r_p = \frac{n_2\cos i - n_1\cos r}{n_2\cos i + n_1\cos r} = \frac{\tan(i-r)}{\tan(i+r)}$$

$$t_s = \frac{2n_1\cos i}{n_1\cos i + n_2\cos r}$$

$$t_p = \frac{2n_1\cos i}{n_2\cos i + n_1\cos r}, \qquad T_{s,p} = \frac{n_2\cos r}{n_1\cos i}|t_{s,p}|^2$$

WATCH / NEVER write T = |t|^2! At air-to-glass normal incidence (n = 1.5), t = 2 / 2.5 = 0.80 -> t^2 = 0.64, whereas TRUE transmitted power is 1.5 x 0.64 = 0.96 (96%)!

Normal incidence, p-sign flip & Stokes relations

At normal incidence R(0) = ((n1 - n2)/(n1 + n2))^2 (4.0% for glass 1.50; 2.0% for water 1.33; 17.2% for diamond 2.42). Reversing rays gives exact Stokes time-reversal laws:

$$r' = -r \quad \text{(time reversal proof of } \pi \text{ flip!)}$$

$$t\,t' + r^2 = 1$$

WHY / Watch r_p closely as i sweeps from 0 to 90 deg on air-glass: r_p crosses ZERO at theta_B = 56.31 deg and flips phase by pi for all grazing angles i > theta_B!

WATCH / As i -> 90 deg (grazing incidence), BOTH R_s -> 100% AND R_p -> 100% with pi phase jump — even rough paper or asphalt acts like a shiny mirror at grazing angle!

Incoherent thick-plate & M-plate pile formulas

For a thick window pane (thickness L >> L_c), multiple internal reflections add INCOHERENTLY as intensities (T0^2 + T0^2 R0^2 + T0^2 R0^4 + ...), NOT amplitudes!

$$T_{1\text{ plate}} = \frac{1-R}{1+R}$$

$$R_{1\text{ plate}} = \frac{2R}{1+R}, \qquad T_{M-\text{plates}} = \frac{1-R}{1+(2M-1)R}$$

Check: single window pane R = 0.04 transmits 0.96 / 1.04 = 92.31% (reflects 7.69%). At theta_B on glass 1.5, R_p = 0 -> T_{M,p} = 100%, while R_s = 25/169 = 14.79%!

WATCH / Distinguish THIN film (L << L_c, coherent sum, oscillates 0 to 4R) from THICK glass window (L >> L_c, incoherent geometric sum, equals 2R/(1+R) smooth)!

Exact quarter-wave AR residual & HR Bragg mirrors

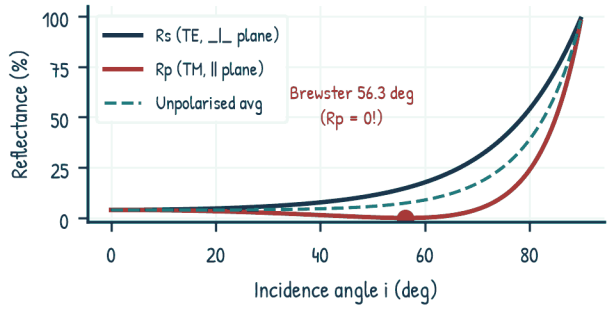
Any quarter-wave dielectric layer (n_c t = lam0 / 4) has exact normal reflectance R_min; alternating N high/low quarter-wave pairs (H/L)^N build 99.99% laser mirrors:

$$R_{\lambda/4} = \left(\frac{n_c^2 - n_1 n_g}{n_c^2 + n_1 n_g}\right)^2, \qquad R_{\text{Bragg},\,N} = \left[\frac{1-(n_g/n_1)(n_H/n_L)^{2N}}{1+(n_g/n_1)(n_H/n_L)^{2N}}\right]^2$$

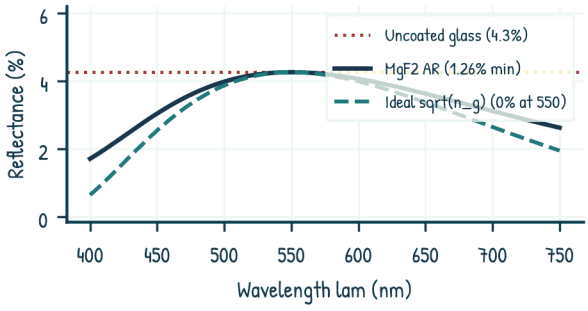
WHY / Single high-index TiO2 quarter-wave layer (n_c = 2.35 > glass 1.52) boosts R UP to 32.3% (lossless beam splitter!), whereas 10 TiO2/SiO2 pairs reach R = 99.98%!

WATCH / Dielectric Bragg mirrors suffer ZERO metallic ohmic heating, letting kilowatt industrial and LIGO cavities handle megawatts of circulating optical power!

Exact air-glass (n = 1.50) Fresnel power reflectances Rs(i) and Rp(i)



Quarter-wave AR spectral curve (550 nm zero explains purple lens tint)



REMEMBER / Reflectance is R = |r|^2, whereas power transmittance carries impedance-projection factor T = (n2 cos r / n1 cos i) |t|^2 = 1 - R!

TIR phase jumps, Fresnel rhomb & evanescent tunnelling

Complex cos(r) beyond critical angle, achromatic Fresnel rhomb QWP, skin decay audit, and frustrated TIR.

TIR unit-modulus reflection & continuous phase

For n1 > n2 (relative n = n2/n1 = sin theta_c < 1) and i > theta_c, cos(r) = i sqrt(sin^2 i / n^2 - 1) makes r_s = exp(-i delta_s) and r_p = exp(-i delta_p) pure phase factors (R = 100%):

$$\tan\left(\frac{\delta_s}{2}\right) = \frac{\sqrt{\sin^2 i - n^2}}{\cos i}$$

$$\tan\left(\frac{\delta_p}{2}\right) = \frac{\sqrt{\sin^2 i - n^2}}{n^2 \cos i}, \qquad \tan\left(\frac{\Delta}{2}\right) = \frac{\cos i \sqrt{\sin^2 i - n^2}}{\sin^2 i}$$

WHY / At i = theta_c both delta_s = delta_p = 0; at i = 90 deg both reach pi; in between, p leads s by peak retardation tan(Delta_max/2) = (1 - n^2)/(2n)!

WATCH / Linearly polarised light striking a TIR prism at azimuth != 0, 90 deg emerges ELLIPTICALLY polarised because delta_p != delta_s!

Fresnel rhomb — achromatic quarter-wave prism

In crown glass (n1 = 1.51 -> n = 0.662), maximum single-bounce retardation is Delta_max = 45.6 deg, hitting exact 45.0 deg at internal angle i = 54.6 deg!

$$\Delta_{2\ \text{bounces}} = 2 \times 45^\circ = 90^\circ = \frac{\pi}{2}$$

$$\Rightarrow 45\ \text{deg linear input} \ \rightarrow\ 100\%\ \text{circular output!}$$

WHY / Unlike crystalline quartz QWP whose phase scales strongly as 1/lam0, Fresnel rhomb depends ONLY on refractive index n(lam), staying ~90 deg across ALL visible colours!

WATCH / Submerging an un-silvered Fresnel rhomb into water raises n = 1.33/1.51 = 0.88, collapsing Delta_max to 7 deg and destroying circular conversion completely!

Evanescent wave & decay-length audit fix (eq. 7.4)

Inside rarer medium 2 (z > 0), field E2 ~ exp(-kappa z) exp(i(k_x x - wt)) skims along x while decaying exponentially along z with ZERO net normal Poynting flux (<S_z> = 0):

$$\kappa = \frac{2\pi}{\lambda_0}\sqrt{n_1^2 \sin^2 i - n_2^2}$$

$$d_E = \frac{1}{\kappa}\ (|E| \propto e^{-z/d_E}), \qquad d_I = \frac{1}{2\kappa}\ (I \propto e^{-2\kappa z} = e^{-z/d_I})$$

WHY / Audit fix (Wave-optics.md eq. 7.4 & Q5): kappa uses 2 pi / lam0 (vacuum lam0!), so intensity/transmission decays as exp(-2 kappa g) = exp(-g / d_I) with 4 pi inside d_I!

WATCH / NEVER put 4 pi inside d AND simultaneously write exp(-2 g / d) — pick ONE notation (exp(-2 kappa g) with kappa = 2 pi/... OR exp(-g / d_I) with d_I = lam0 / 4 pi...)!

Frustrated TIR tunnelling & Goos-Hanchen shift

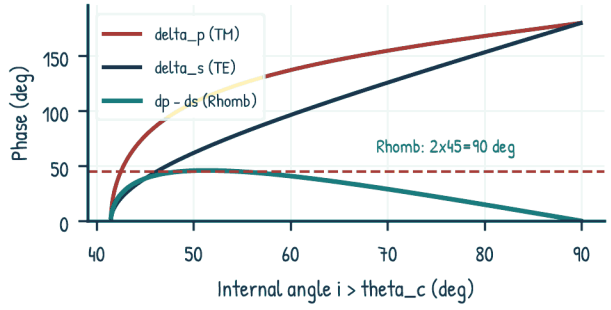
Bringing second glass prism n1 within sub-wavelength gap g (~lam0) couples the evanescent tail back into a propagating wave — exact optical twin of quantum alpha-tunnelling!

$$T_{\text{FTIR}}(g) = \frac{1}{1 + C_{s,p}\sinh^2(\kappa g)} \approx \frac{4}{C_{s,p}} e^{-2\kappa g} \quad (\kappa g \gg 1)$$

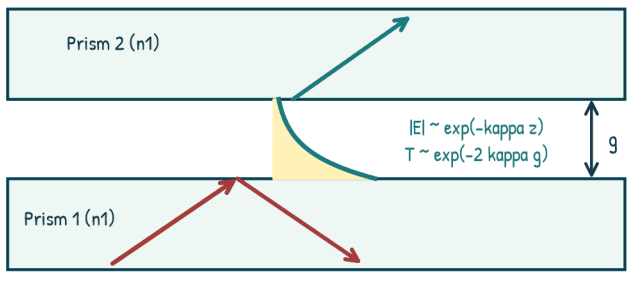
WHY / Touching skin ridges to a TIR prism frustrates air gap g -> 0 along ridges while valleys keep g >> d_E, imaging fingerprints sharply; TIRF microscopy illuminates only 80 nm of cells!

WATCH / A narrow laser beam undergoing TIR also steps laterally forward along the boundary by Goos-Hanchen shift dx_GH = -d(delta)/dk_x ~ O(lam0)!

TIR phase shifts in crown glass: two 45-deg bounces make an achromatic QWP



Evanescent field skin depth and Frustrated TIR optical barrier tunnelling



REMEMBER / Keep amplitude decay length d_E = 1/kappa SEPARATE from intensity length d_I = 1/(2 kappa) so factor 2 is NEVER double-counted!

Exact N-slit polygon, blazed echelle & acousto-optic gratings

Exact (sin N gamma / sin gamma)^2 factor, subsidiary peaks, oblique Littrow gratings, and ultrasonic phase gratings.

Exact N-slit phasor sum & subsidiary peaks

Summing N equal slits of width a and spacing d gives geometric series sum_{q=0}^{N-1} exp(i 2 q gamma) = exp(i(N-1)qamma) sin(N qamma)/sin(qamma):

$$I(\theta) = I_1(0)\left(\frac{\sin(N\gamma)}{\sin\gamma}\right)^2\left(\frac{\sin\beta}{\beta}\right)^2 \qquad \left(\gamma = \frac{\pi d \sin\theta}{\lambda},\quad \beta = \frac{\pi a \sin\theta}{\lambda}\right)$$

WHY / Principal peaks hit N^2 I1 with half-width dgamma = pi/N (~1/N width), conserving total energy N^2 x (1/N) = N; first side lobe sits at ~1/(3 pi / 2)^2 = 4.5% of peak!

WATCH / Count carefully: between consecutive principal maxima lie N - 1 dark zeros (N gamma = p pi, p not multiple of N) and N - 2 weak subsidiary maxima!

Oblique grating equation, minimum deviation & Littrow

With incident ray at angle i and diffracted ray at theta_m on the SAME side of normal (reflection convention), path difference across qroove step d is d(sin i + sin theta_m):

$$d(\sin i + \sin\theta_m) = m\lambda$$

$$m_{\max}^{\text{oblique}} = \left\lfloor \frac{2d}{\lambda} \right\rfloor, \qquad 2d\sin\left(\frac{D_{\min}}{2}\right) = m\lambda$$

WHY / In auto-collimating Littrow mount (diffracted ray returns straight back along incident ray, theta_m = i!), 2 d sin(i) = m lam gives R = mN = 2 W sin(i) / lam!

WATCH / Normal incidence caps orders at floor(d/lam), whereas grazing incidence + grazing diffraction doubles reach to floor(2d/lam)!

Free spectral range & sawtooth blazed echelles

Unblazed gratings dump most energy into useless zero order m = 0. Tilting reflective groove facets by blaze angle theta_b steers specular reflection straight onto target order m!

$$\Delta\lambda_{\text{FSR}} = \frac{\lambda}{m}, \qquad \lambda_{\text{blaze}}^{\text{Littrow}} = \frac{2d\sin\theta_b}{m}$$

WHY / Astronomical echelle spectrographs run steep facets (theta_b ~ 63.4 deg) at orders m = 50-100 for R > 100,000, using a cross-disperser prism to separate stacked orders!

WATCH / Higher order m sharpens resolution (R = mN) while squeezing Free Spectral Range (lam / m) — exact same tradeoff as Fabry-Perot etalons!

2D meshes, optical discs & ultrasonic gratings

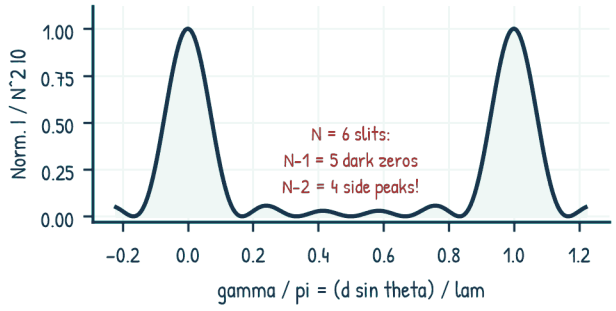
Crossed 2D mesh (d_x, d_y) forms reciprocal grid spots (p lam / d_x, q lam / d_y). In water, an ultrasonic wave of speed v_s and frequency f_s creates refractive index ripples:

$$d_{\text{acoustic}} = \lambda_{\text{sound}} = \frac{v_s}{f_s} \Rightarrow \lambda_{\text{sound}}\sin\theta_m = m\lambda_{\text{light}}$$

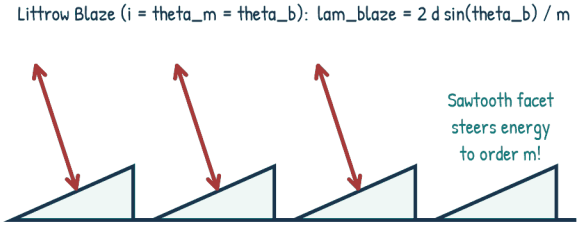
WHY / Debye-Sears acousto-optic modulators steer laser beams electronically in microseconds simply by tuning RF drive frequency f_s!

WATCH / Instantaneous refractive-index grating period equals FULL acoustic wavelength lam_sound = v_s / f_s (NOT lam_sound / 2)!

Exact N-slit interference factor sharpening principal lines as width ~ 1/N



Sawtooth blazed reflection echelle grating concentrates 85% power into order m



REMEMBER / Between any two adjacent principal grating peaks sit STRICTLY N - 1 dark zeros and N - 2 subsidiary peaks; maximum oblique order is floor(2d / lam)!

Fabry-Perot etalons, Airy finesse & laser cavities

Infinite multiple-beam geometric series, 100% resonant transmission miracle, finesse vs F, and Wiener standing wave.

Exact Airy formula & 100% resonance miracle

Two parallel lossless mirrors of reflectance R = |r|^2 spaced by n L sum geometric series T0 / (1 - R exp(i delta)) with round-trip phase delta = 4 pi n L cos(theta) / lam0:

$$T_{\mathrm{FP}}(\delta) = \frac{1}{1 + F\sin^2(\delta/2)}$$

$$R_{\mathrm{FP}}(\delta) = \frac{F\sin^2(\delta/2)}{1 + F\sin^2(\delta/2)}, \qquad F \equiv \frac{4R}{(1-R)^2}$$

WHY / Even with TWO 99% mirrors, on resonance (delta = 2m pi) leakage from giant intracavity power 1/(1 - R) cancels front reflection destructively, giving T_FP = 100%!

WATCH / Mirror absorption A > 0 (R + T0 + A = 1) spoils cancellation and caps peak transmission at T_max = [T0 / (T0 + A)]^2!

Coefficient F vs Finesse F_cal (notation alert!)

Resonances repeat every 2 pi in phase while each Lorentzian-like spike has full width at half maximum ddelta_FWHM = 4 / sqrt(F) = 2(1 - R) / sqrt(R):

$$\mathcal{F} \equiv \frac{2\pi}{\delta_{\mathrm{FWHM}}} = \frac{\pi\sqrt{R}}{1-R} = \frac{\pi\sqrt{F}}{2}$$

$$(R = 0.96 \Rightarrow \mathcal{F} = 77.0, \quad F = 3840)$$

WHY / Finesse F_cal counts how many round-trip beams interfere constructively inside the cavity — exact multiple-beam analogue of grating slit count N!

WATCH / NEVER plug F = 4R/(1 - R)^2 into R = m F_cal! Chromatic resolving power uses FINESSE pi sqrt(R)/(1 - R), which is smaller by ~sqrt(F)!

Free spectral range & hyperfine resolving power

Normal resonance m lam = 2 n L separates longitudinal modes by Free Spectral Range c / (2 n L) and achieves ultra-high chromatic resolving power m * F_cal:

$$\Delta\nu_{\mathrm{FSR}} = \frac{c}{2nL}$$

$$\Delta\lambda_{\mathrm{FSR}} = \frac{\lambda^2}{2nL}, \qquad \mathcal{R} = \frac{\lambda}{\delta\lambda_{\min}} = m\,\mathcal{F} = \frac{2nL}{\lambda}\,\frac{\pi\sqrt{R}}{1-R}$$

Anchor: L = 1 cm air etalon, R = 0.97 (F_cal = 103), lam = 500 nm -> order m = 40,000 -> R = 4.1e6 (resolves 0.12 pm Zeeman/hyperfine splittings)!

WATCH / He-Ne laser (gain width 1.5 GHz, cavity L = 30 cm -> FSR = 0.50 GHz) oscillates on ~3 simultaneous longitudinal modes spaced by 500 MHz!

Wiener 1890 standing light wave experiment

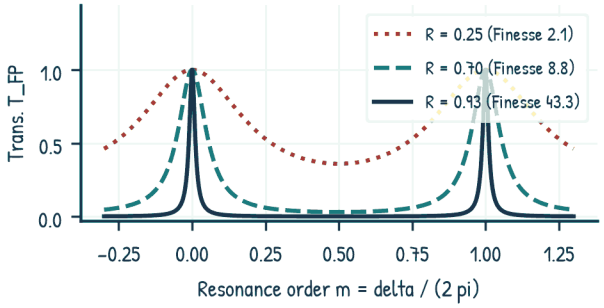
Normal silver-mirror reflection r = -1 forms optical standing wave |E(z)| = 2 E0 |sin(kz)| with E-nodes at m lam/2 and B-antinodes shifted by lam/4 (B maximum AT z = 0!):

$$z_{E-\mathrm{antinode}} = \left(m + \tfrac{1}{2}\right)\tfrac{\lambda}{2}, \qquad D_{\mathrm{emulsion\ bands}} = \frac{\lambda}{2\sin\alpha}$$

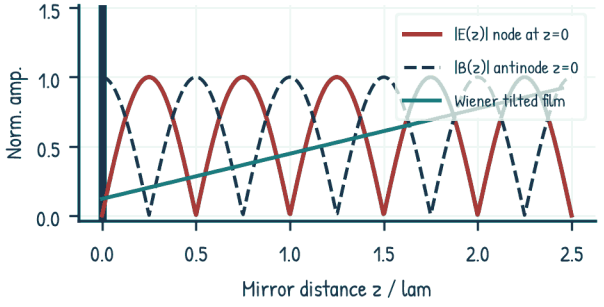
WHY / Wiener sliced the standing field with a 20 nm film tilted by alpha ~ 1e-3 rad: zero blackening at mirror z = 0 proved silver grains respond to ELECTRIC vector E, not B!

WATCH / In fiber-optic Bragg sensors and Lippmann colour plates, standing waves write refractive layers spaced by lam / (2 n), reflecting that exact wavelength back!

Fabry-Perot Airy resonances reach T = 100% even when each mirror reflects 99%



Wiener 1890 experiment: emulsion blackens at E antinodes, proving E acts on matter



REMEMBER / Distinguish coefficient F = 4R/(1 - R)^2 from reflective finesse F_cal = pi sqrt(R)/(1 - R); resolving power equals R = m * F_cal!

Fresnel half-period zones, zone plates & Cornu spiral

Equal-area concentric zones, A1/2 spiral convergence, Arago bright spot, Babinet principle, and straight-edge I0/4.

Fresnel half-period zones & 4 I0 aperture miracle

Divide plane wavefront at distance b from axial point P into rings whose outer radii r_m add path m lam / 2: EVERY zone has equal area pi lam b and opposite sign!

$$r_m = \sqrt{m\lambda b} \quad \left(\text{or } \sqrt{\tfrac{m\lambda ab}{a+b}}\right)$$

$$A_{\text{open}} = A_1 - A_2 + A_3 - \ \ldots = \tfrac{A_1}{2} \Rightarrow I_{1\ \text{zone}} = 4I_0$$

WHY / Opening ONLY zone 1 blocks zone 2 destructive cancellation, making axial spot P FOUR TIMES brighter than removing the entire opaque screen!

WATCH / Widening hole to include 2 zones gives A1 - A2 ~ 0 (DARKNESS on axis!); odd number of zones -> ~4 I0 bright, even number -> ~0 dark!

Arago-Poisson bright spot & Babinet principle

A smooth coin/ball blocking the first k zones leaves A_{k+1} - A_{k+2} + ... = A_{k+1} / 2 ~ A1 / 2 -> UNOBSTRUCTED brightness I0 right at the dead centre of its shadow!

$$E_{\text{aperture}}(P) + E_{\text{obstacle}}(P) = E_{\text{open}}(P)$$

$$\Rightarrow I_{\text{wire}}(\theta \neq 0) = I_{\text{slit}}(\theta \neq 0)$$

WHY / Babinet principle equates diffraction away from theta = 0 for ANY complementary aperture/obstacle pair — how laser micrometers measure human hair thickness w via w sin(theta) = m lam!

WATCH / Arago spot requires a smooth circular rim and spatially coherent illumination — rough jagged coin edges scramble zone phases and erase the central spot!

Fresnel zone plate — flat multi-focus X-ray lens

Blocking all even zones lets all M open odd zones add IN PHASE at primary focal length f1 = r1^2 / lam AND subsidiary odd foci f1/3, f1/5, ... (plus diverging virtual foci -f_k!):

$$f_k = \frac{r_1^2}{(2k-1)\lambda}$$

$$I_{\text{focus, amp}} = 4M^2 I_0, \qquad I_{\text{focus, phase plate}} = 16M^2 I_0$$

WHY / Etching even zones to phase depth t = lam0 / [2(n - 1)] flips -A_{2m} to +A_{2m}, doubling amplitude again andboosting focal intensity 4x over black-ring plates!

WATCH / Zone-plate focal length varies inversely with wavelength (f1 ~ 1/lam: red focuses CLOSER than blue — opposite to refractive glass lenses)!

Cornu clothoid spiral & straight-edge shadow law

Fresnel integrals C(v), S(v) coil between eyes (-1/2, -1/2) and (+1/2, +1/2) (open chord squared = 2 -> I0). A semi-infinite razor edge blocks v < 0:

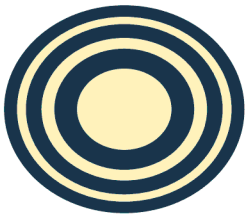
$$I_{\text{geometric edge}}(v=0) = \tfrac{I_0}{2}\left[\left(\tfrac{1}{2}\right)^2 + \left(\tfrac{1}{2}\right)^2\right] = \mathbf{\tfrac{I_0}{4}}$$

$$I_{\text{1st bright}} = 1.370\, I_0$$

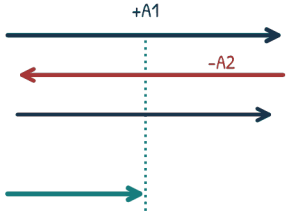
WHY / Cutting half the wavefront halves ELECTRIC FIELD AMPLITUDE to A0 / 2, so irradiance on the geometric shadow line equals (1/2)^2 I0 = I0 / 4 = 25%!

WATCH / Inside the shadow of a single straight edge intensity decays monotonically to zero with ZERO dark fringes; dark fringes appear ONLY on the lit side!

Fresnel half-period zone plate and alternating spiral sum A1 - A2 + A3 - ... = A1/2

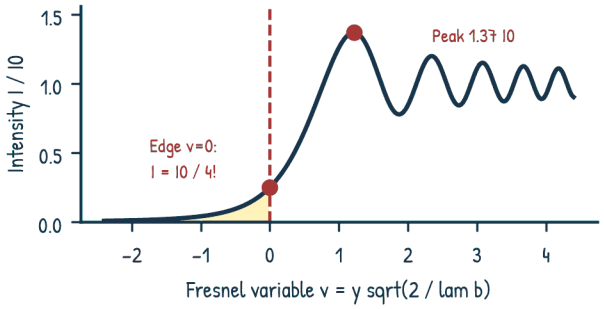


Zone radii r_m = sqrt(m lam b)



A_open = A1/2 => 1-zone I = 4 I0!

Fresnel straight-edge shadow transition: exact I0/4 at geometric edge v = 0



REMEMBER / 1 open Fresnel zone gives 4 I0 (2 zones give ZERO!); right on a straight razor shadow edge intensity is I0 / 4, NEVER I0 / 2!

Abbe Fourier imaging, X-ray Bragg law, Sagnac & HBT

Back-focal-plane spatial filtering, Zernike phase contrast, crystal Bragg glancing law, fiber gyros, and HBT bunching.

Abbe spatial filtering & Zernike phase contrast

Objective lens forms spatial Fourier transform of specimen at back focal plane F-prime (spot spacing x_m = m lam f / Lambda). Image periodicity reconstructs iff >= 2 orders pass:

$$\Lambda_{\min}^{\text{axial coherent}} = \frac{\lambda_0}{\mathrm{NA}}, \qquad \Lambda_{\min}^{\text{oblique cone}} = \frac{\lambda_0}{2\,\mathrm{NA}}$$

WHY / Blocking DC spot m = 0 in F-prime yields Dark-Field imaging; retarding m = 0 by pi/2 converts invisible cell phase shifts phi(x) into direct intensity contrast 1 +/- 2 phi(x)!

WATCH / Putting a vertical slit mask inside F-prime passes horizontal diffraction orders and erases horizontal lines while leaving vertical specimen lines sharp!

3D X-ray Bragg crystal reflection & n < 1 shift

Crystal lattice planes spaced by d_{hkl} = a0 / sqrt(h^2 + k^2 + l^2) reflect X-rays at GLANCING angle theta when adjacent-plane path difference 2 d sin(theta) equals n lam:

$$2d_{hkl}\sin\theta = n\lambda \quad (\lambda \le 2d_{hkl})$$

$$n_{\text{X ray}} = 1 - \delta \Rightarrow 2d\sin\theta_n\left(1 - \frac{\delta}{\sin^2\theta_n}\right) = n\lambda$$

WHY / Because X-ray frequencies sit ABOVE atomic electron resonances, refractive index n = 1 - delta < 1 (delta ~ 10^(-5)), causing external grazing TIR below theta_c = sqrt(2 delta)!

WATCH / NUMBER-ONE X-RAY TRAP: if X-rays hit at 60 deg TO THE SURFACE NORMAL, glancing Bragg angle is theta = 90 - 60 = 30 deg (and beam deflection is 2 theta = 60 deg)!

Sagnac rotating ring & Fresnel-drag cancellation

Counter-propagating beams around rotating loop area A (angular speed Omega, N fiber turns) experience opposite beam-splitter motion, accumulating Sagnac phase shift:

$$\Delta t = \frac{4N\Omega\cdot\mathbf{A}}{c^2}, \qquad \Delta\Phi_{\text{Sagnac}} = \frac{8\pi N\Omega\cdot\mathbf{A}}{\lambda_0 c}, \qquad \Delta\nu_{\text{RLG}} = \frac{4\Omega\cdot\mathbf{A}}{\lambda_0 P}$$

WHY / Relativistic Fresnel drag v = c/n +/- R Omega (1 - 1/n^2) inside co-moving silica fiber cancels n^2 identically, leaving Delta_Phi INDEPENDENT of refractive index n!

WATCH / IPhO/APhO favourite red herring: stem gives n = 1.46 silica fiber and asks for new Sagnac shift vs air loop — answer is UNCHANGED (1 : 1)!

Hanbury Brown-Twiss (HBT) & 3D holography

HBT correlates intensity fluctuations <I1(t) I2(t + tau)> electrically between separated telescopes; thermal photon bunching locks g^(2)(tau) directly to |gamma_12|^2:

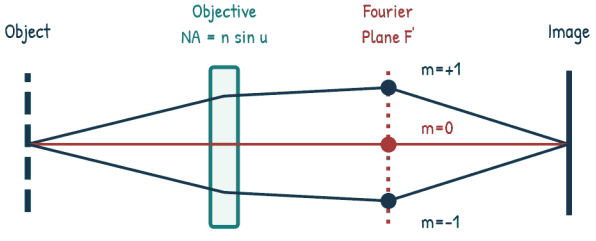
$$g^{(2)}(\tau) = \frac{\langle I_1(t)\, I_2(t+\tau)\rangle}{\langle I_1\rangle\langle I_2\rangle} = 1 + |\gamma_{12}(\tau)|^2$$

$$I_{\text{holo}} = |E_r + E_o|^2 = I_r + I_o + E_r^*\,E_o + E_r E_o^*$$

WHY / Phase turbulence cancels out of |gamma_12|^2 in HBT; in off-axis holography, interference fringes encode BOTH amplitude and phase of E_o onto flat film!

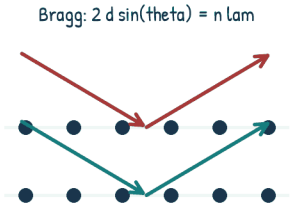
WATCH / Cutting a hologram into quarters does NOT chop the 3D image into quarters — EVERY piece still reconstructs the ENTIRE 3D object with slightly lower resolution!

Abbe two-step Fourier imaging and spatial filtering in back focal plane F-prime



Need >= 2 orders inside aperture => d_min = lam0 / (2 NA)

Crystal X-ray Bragg reflection (glancing angle!) and rotating Sagnac interferometer



theta measured FROM plane!



dPhi = 8 pi Omega.A / (lam0 c)

REMEMBER / Bragg angle theta is ALWAYS measured from the crystal plane (NOT normal!); fiber Sagnac phase shift is 100% independent of core index n!